Practical Predictive AI Evaluation

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Duke B&B

Overview

We'll discuss important considerations relevant to evaluating AI tools and making sense of their predictions in clinical/healthcare practice:

- How might we use AI-based predictions, and what performance metrics are relevant to each use case?
  - Discrimination metrics
  - Calibration metrics

- What information about the model do we need to know to use it responsibly?

Predictive AI

Use cases and associated performance metrics

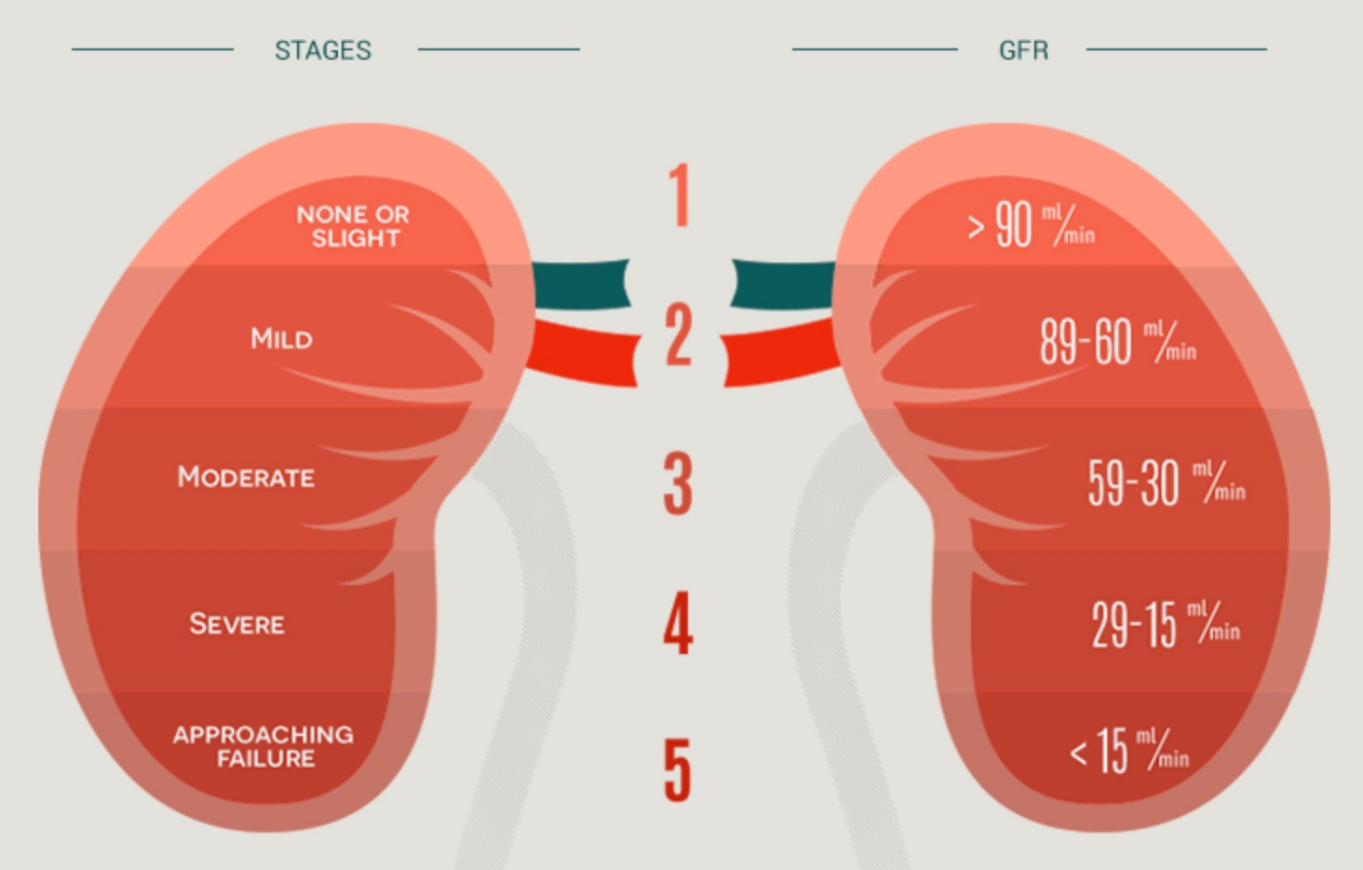
New Stage IV Risk Prediction Model

Some of your colleagues have implemented a new prediction model that aims to prevent or delay kidney failure by predicting 2-year risk of progression to stage IV CKD.

This model is based on electronic health record data, including diagnosis and procedure codes, medications, labs, and vitals; as well as social determinants of health.

You think the model sounds promising and start thinking about ways you might use it in practice.

Prediction: Probability of a Binary Outcome



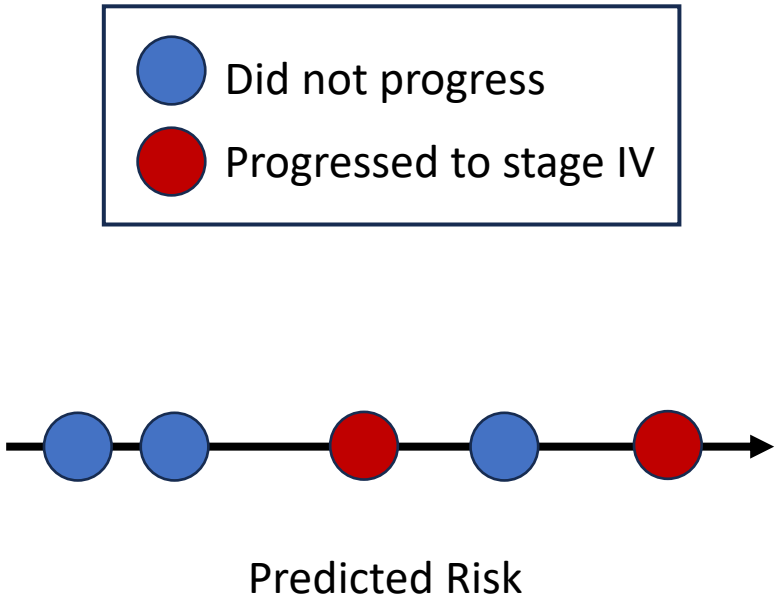
11%

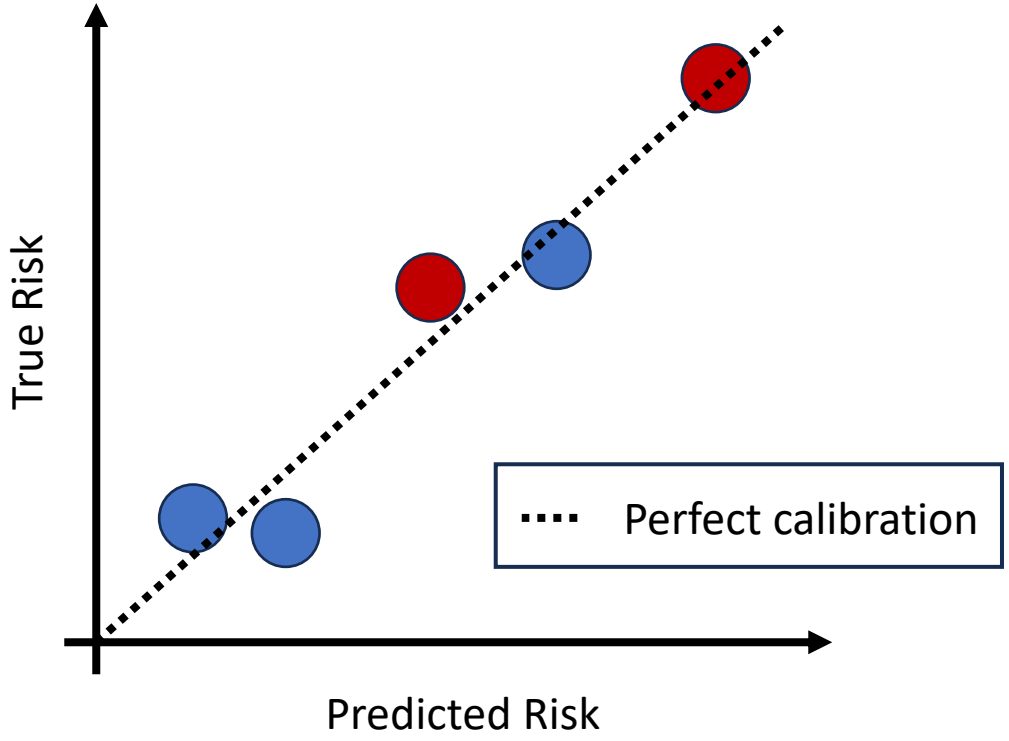
Probability of Stage IV CKD within two years

<u>Report Back</u>:

- How might you use this number in clinical practice (be specific)?
- What performance metrics are relevant to this way of using the prediction?

Discrimination vs Calibration





Can we identify individuals who will progress?

Are our predicted probabilities accurate?

Assessing model discrimination

Sensitivity, specificity, PPV, NPV (threshold-based)

AUC, AUPRC (threshold-agnostic)

Suppose our features are highly informative. What might our model's predictions look like?

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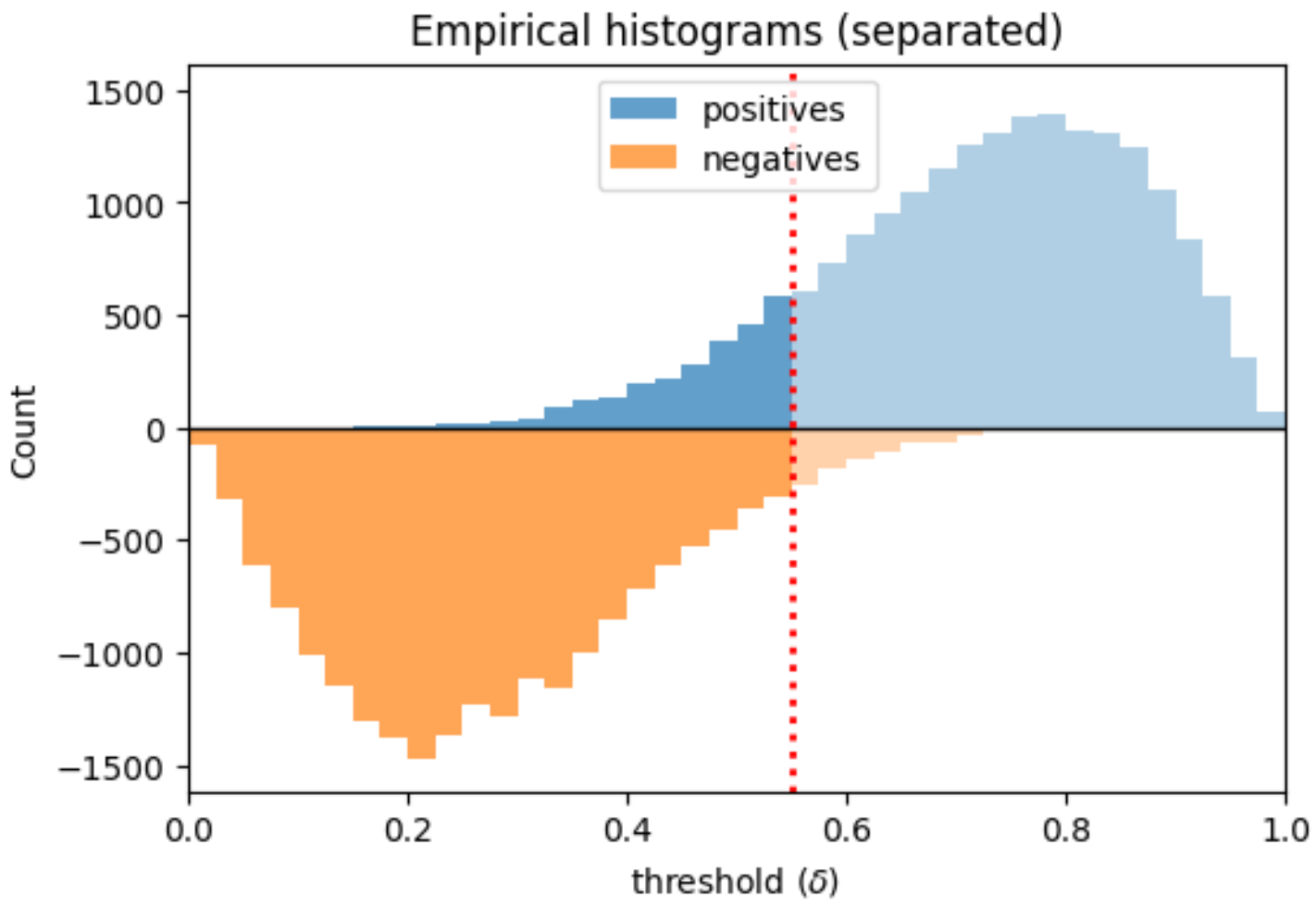

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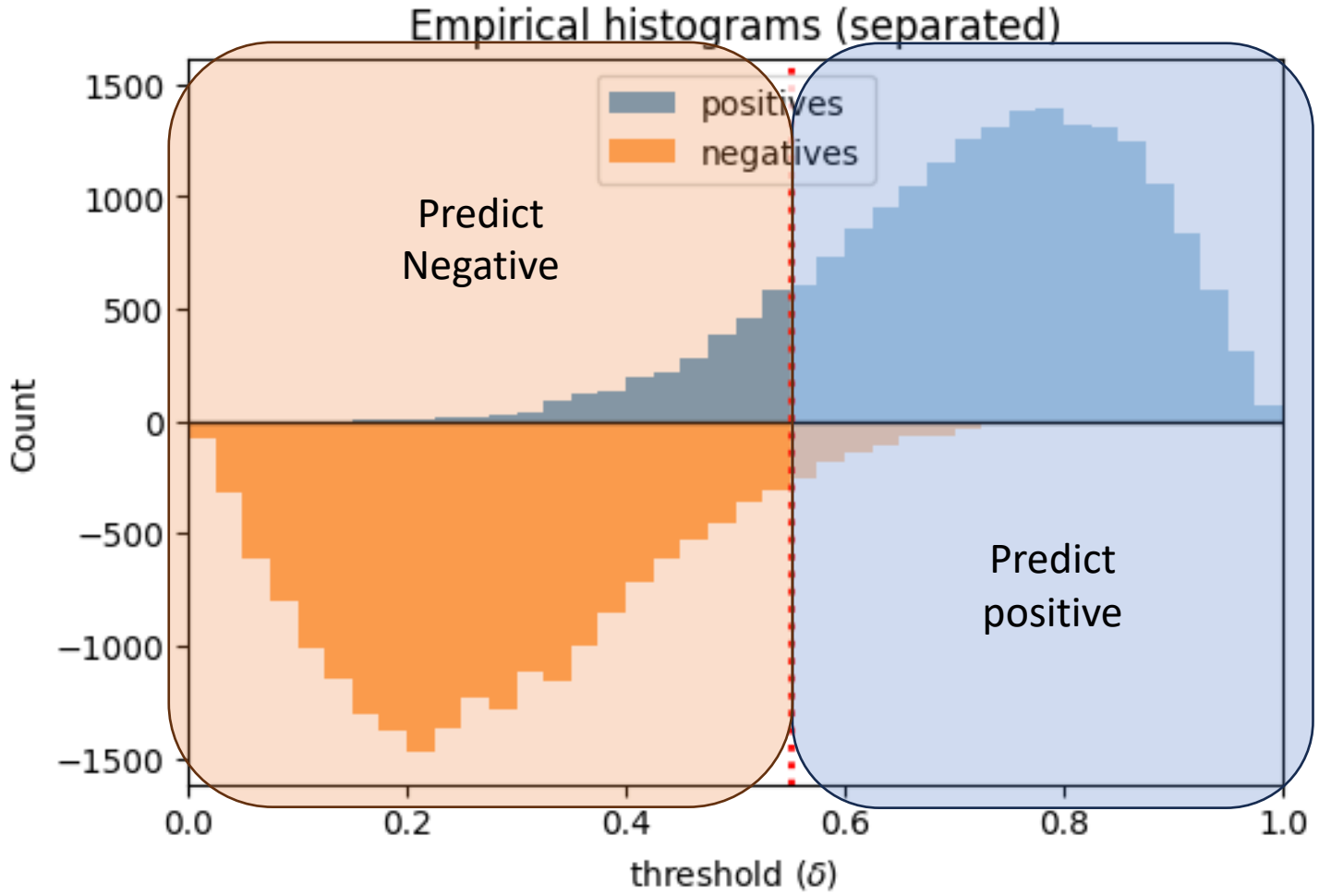

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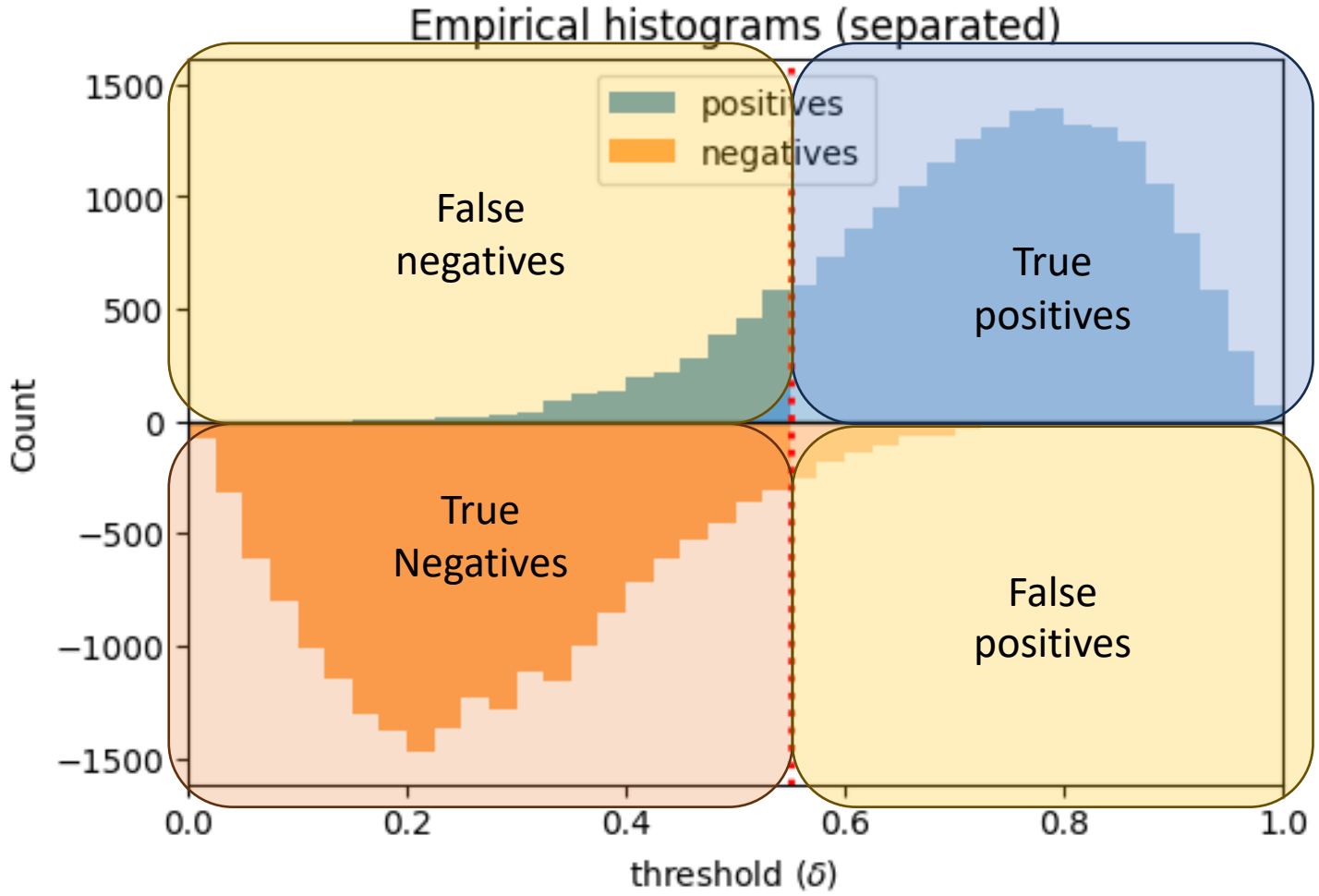

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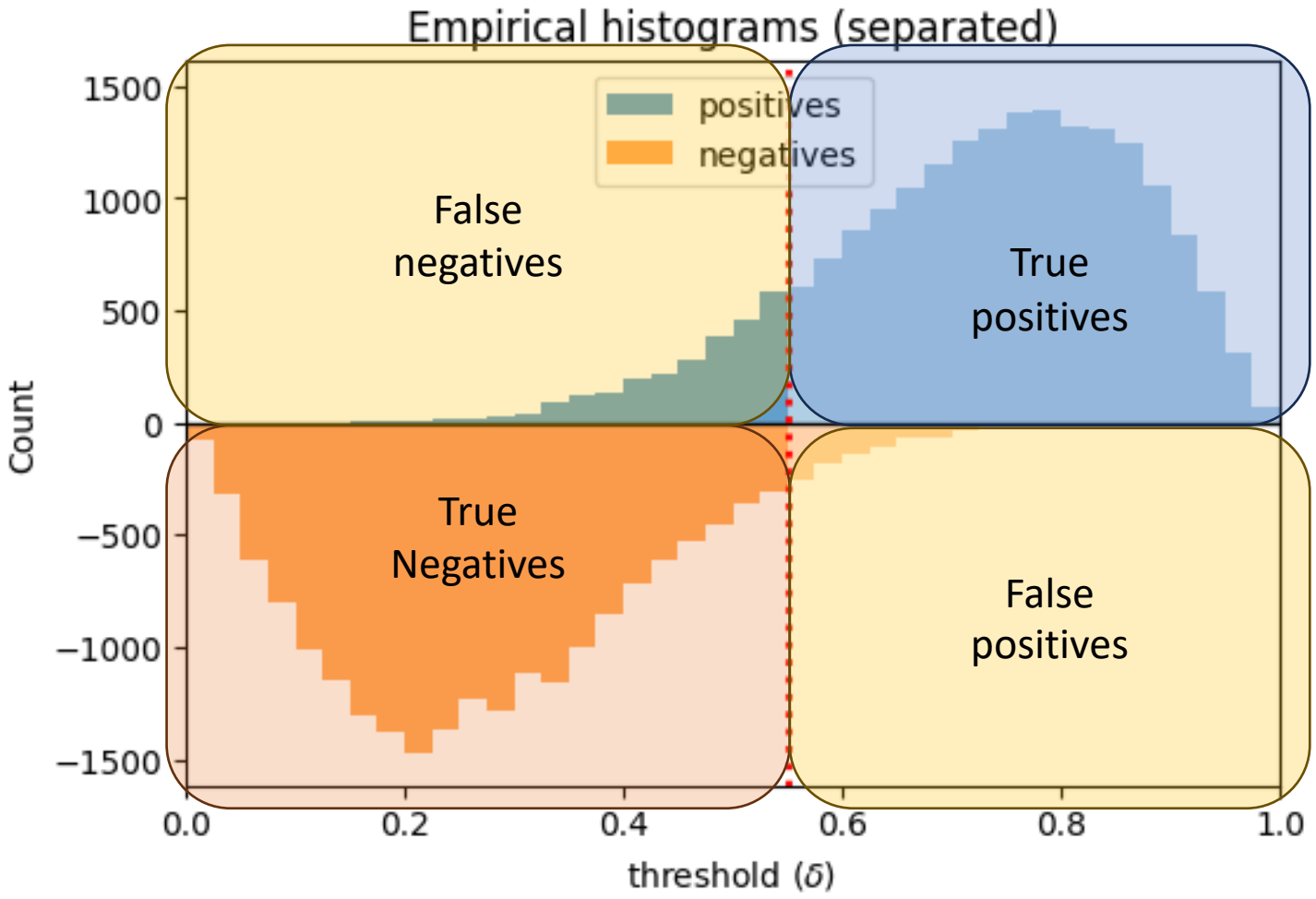
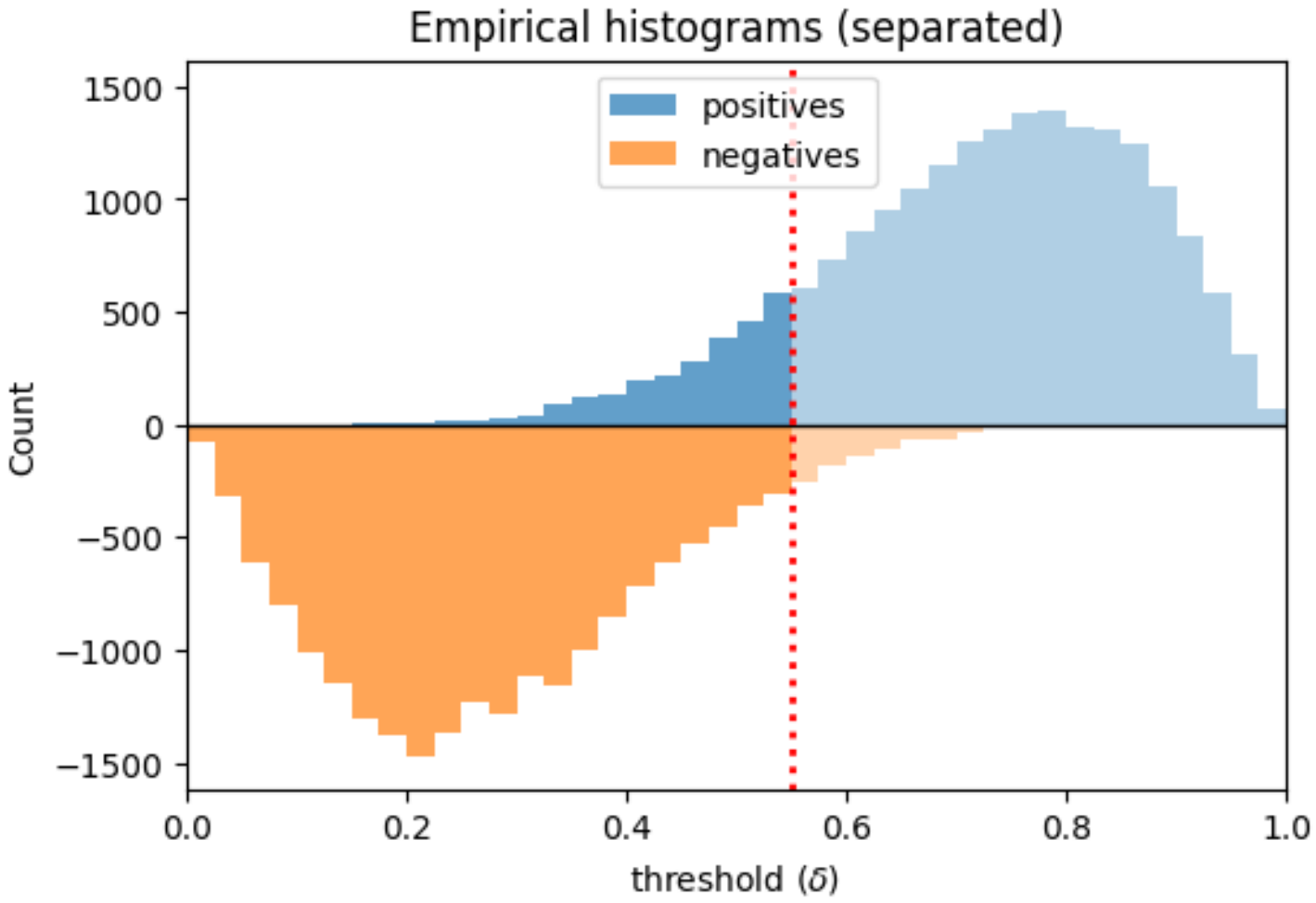


Review threshold-specific metrics:

- TPR (sensitivity, recall)
- FPR (1 – specificity)
- Precision (PPV)
- NPV

Consider the clinical utility of each!!

Consider: how does each metric change as we vary the threshold?



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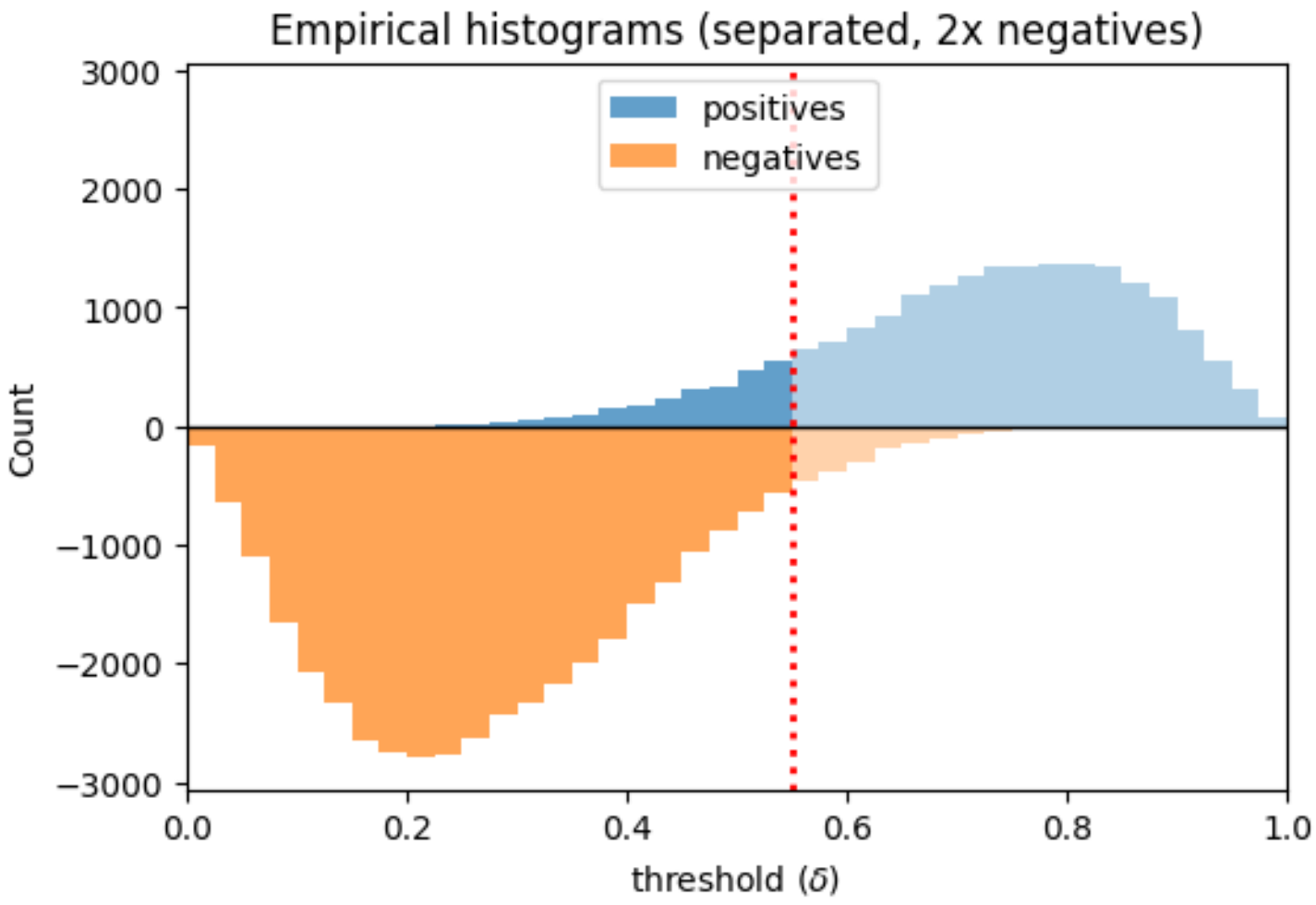
Consider: how does each metric change as we vary class balance?



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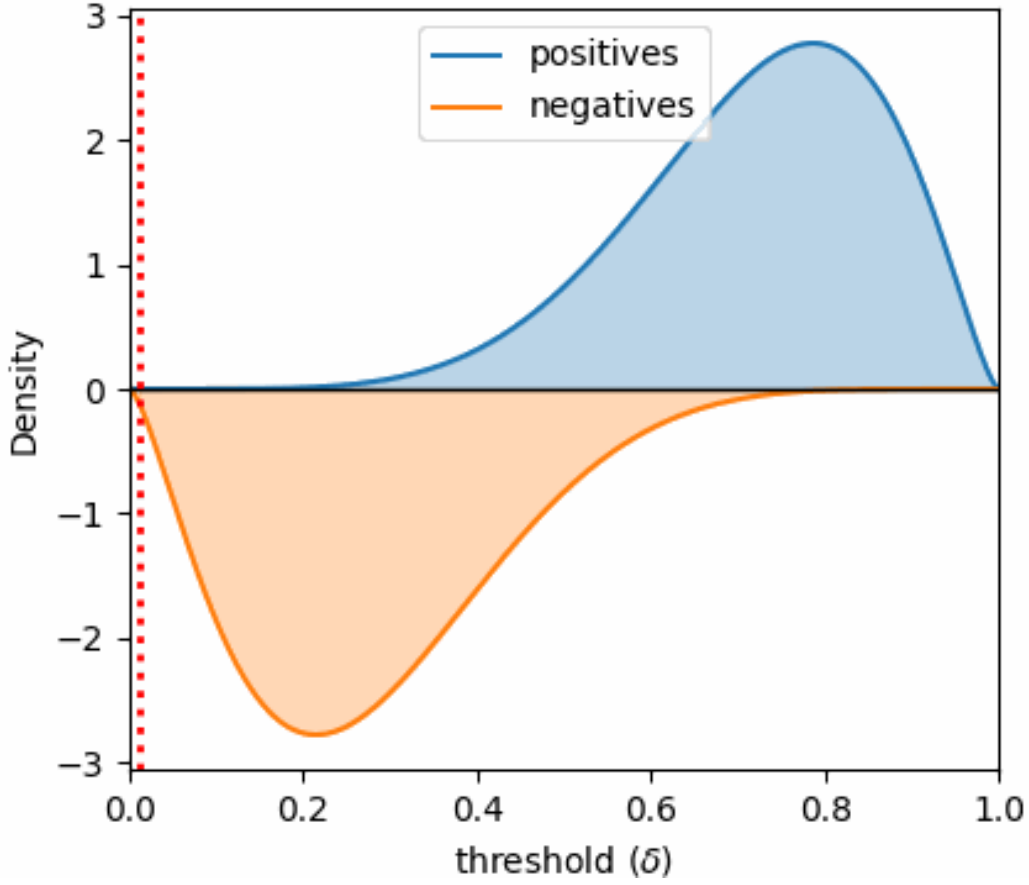
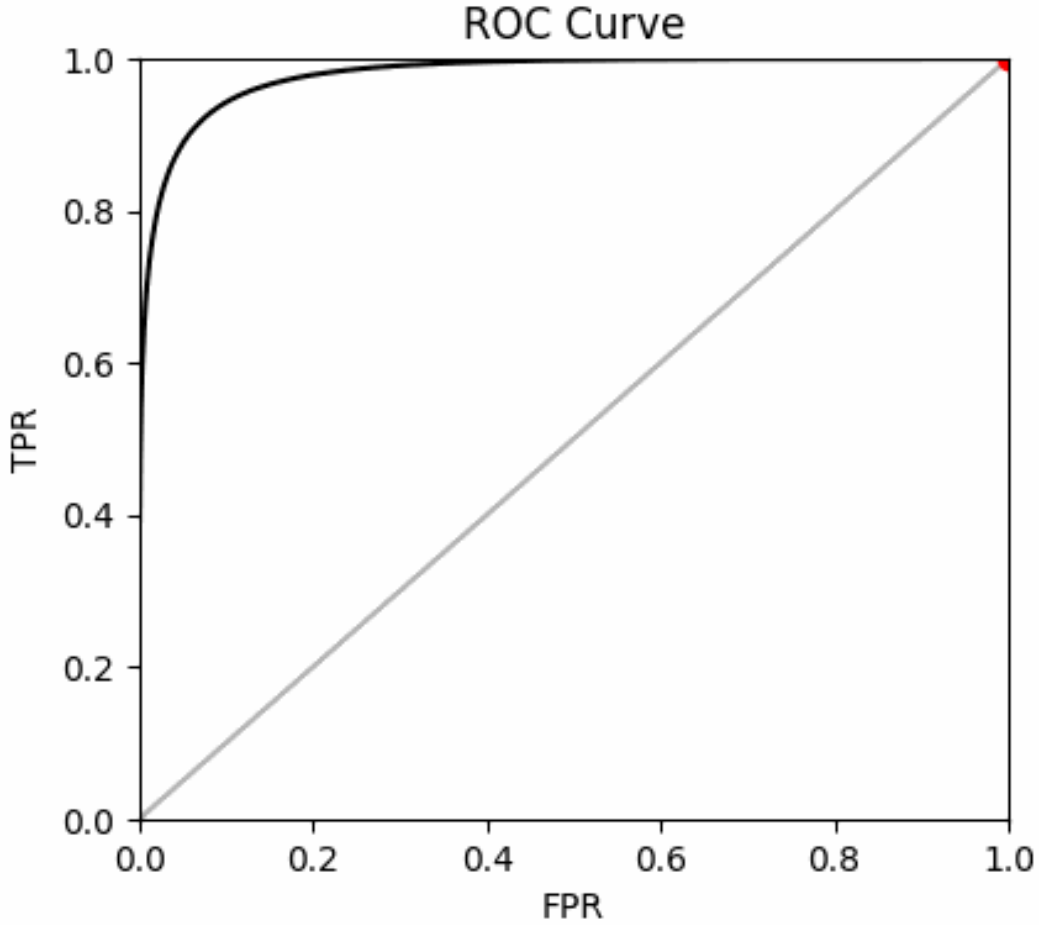
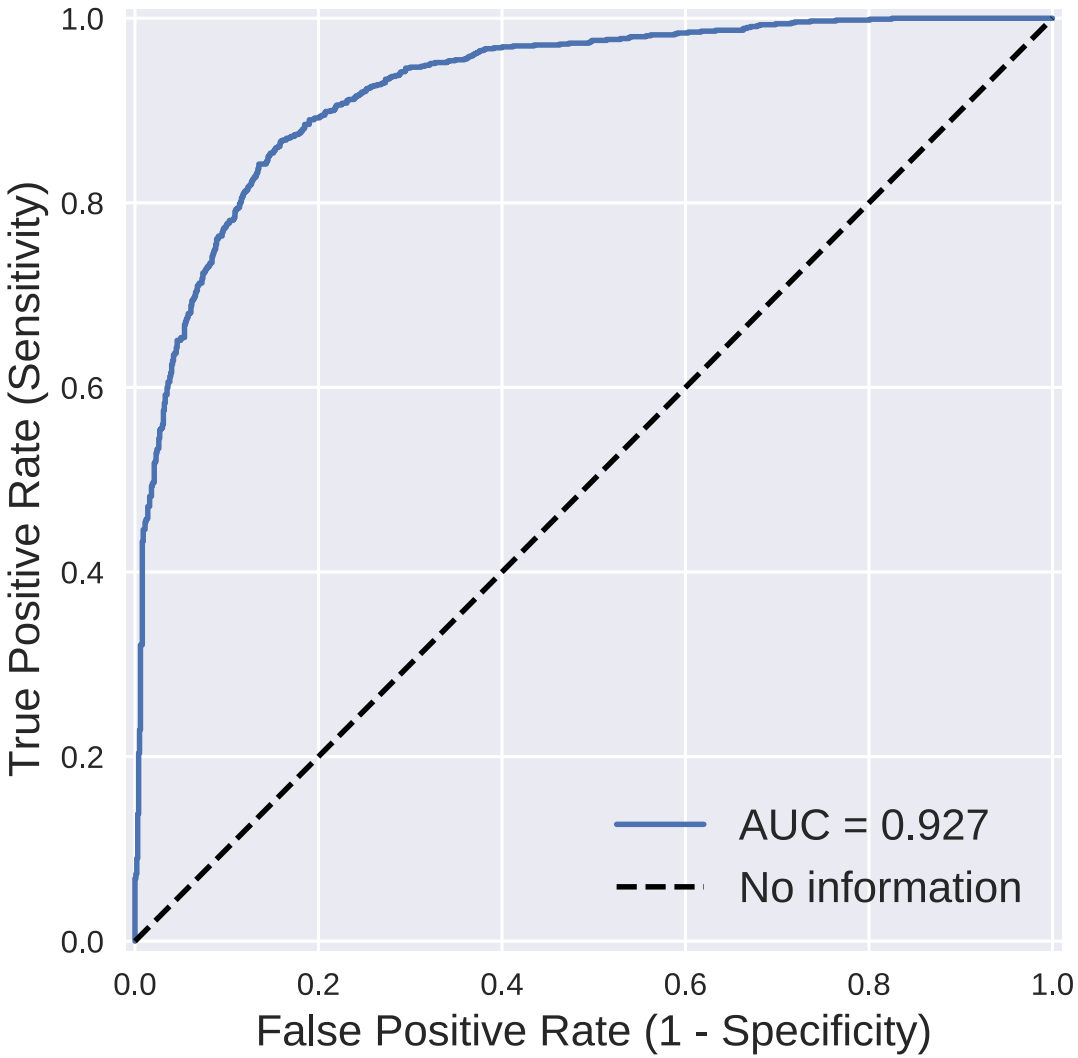




Receiver Operating Characteristic Curve

- Illustrates the tradeoff between the TPR (sensitivity) and FPR (1–specificity) as we vary the threshold.
- Allows clinicians to consider this tradeoff and select appropriate operating points
- The area under this curve (AUC) provides a single number summarizing this tradeoff.
- <u>In most cases this is the most important metric (why??)</u>

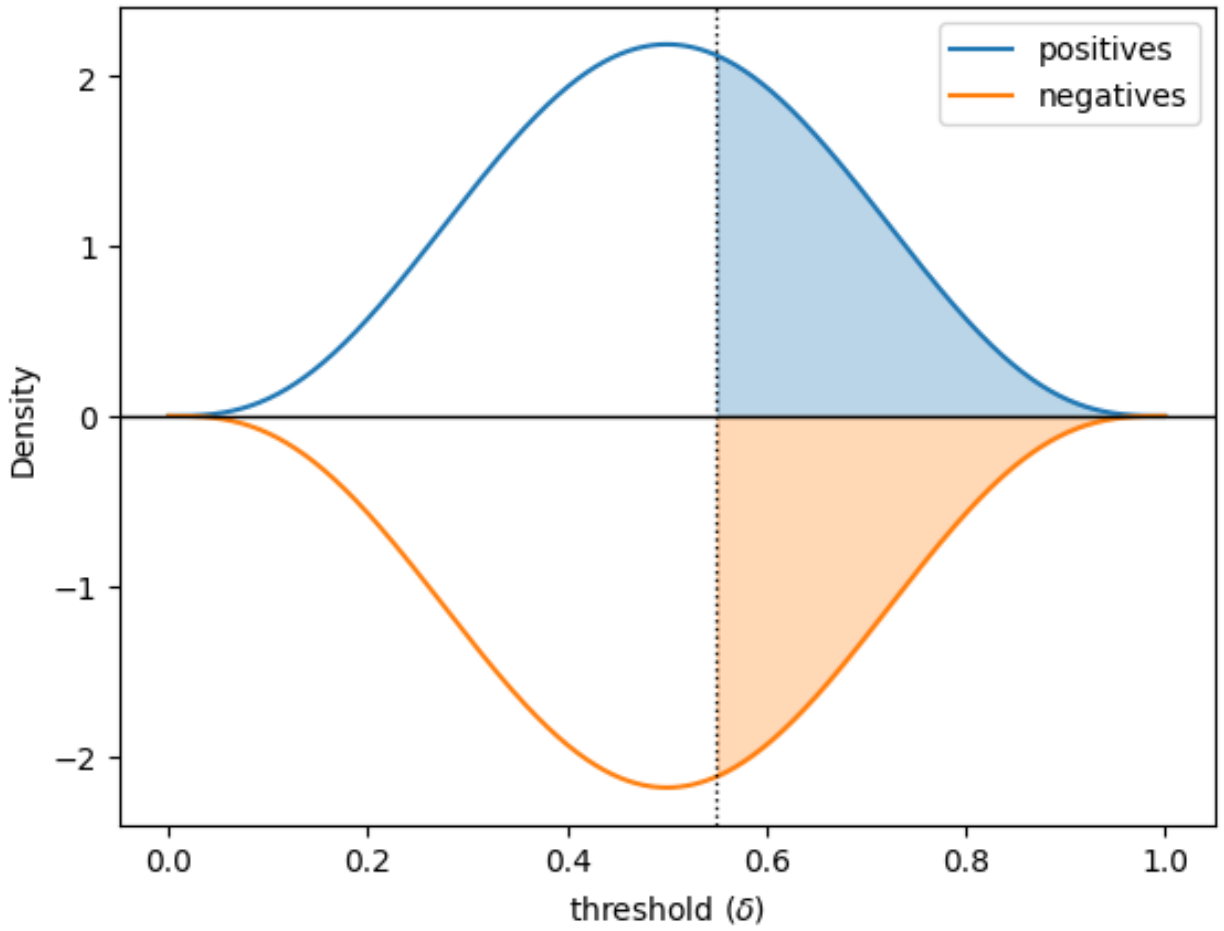


So, what's a *good* AUC value?
(i.e., *good* performance)?

We'll start to answer this question by taking a look at *bad* performance.

Suppose our features contain no information about the outcome. What might predictions look like?

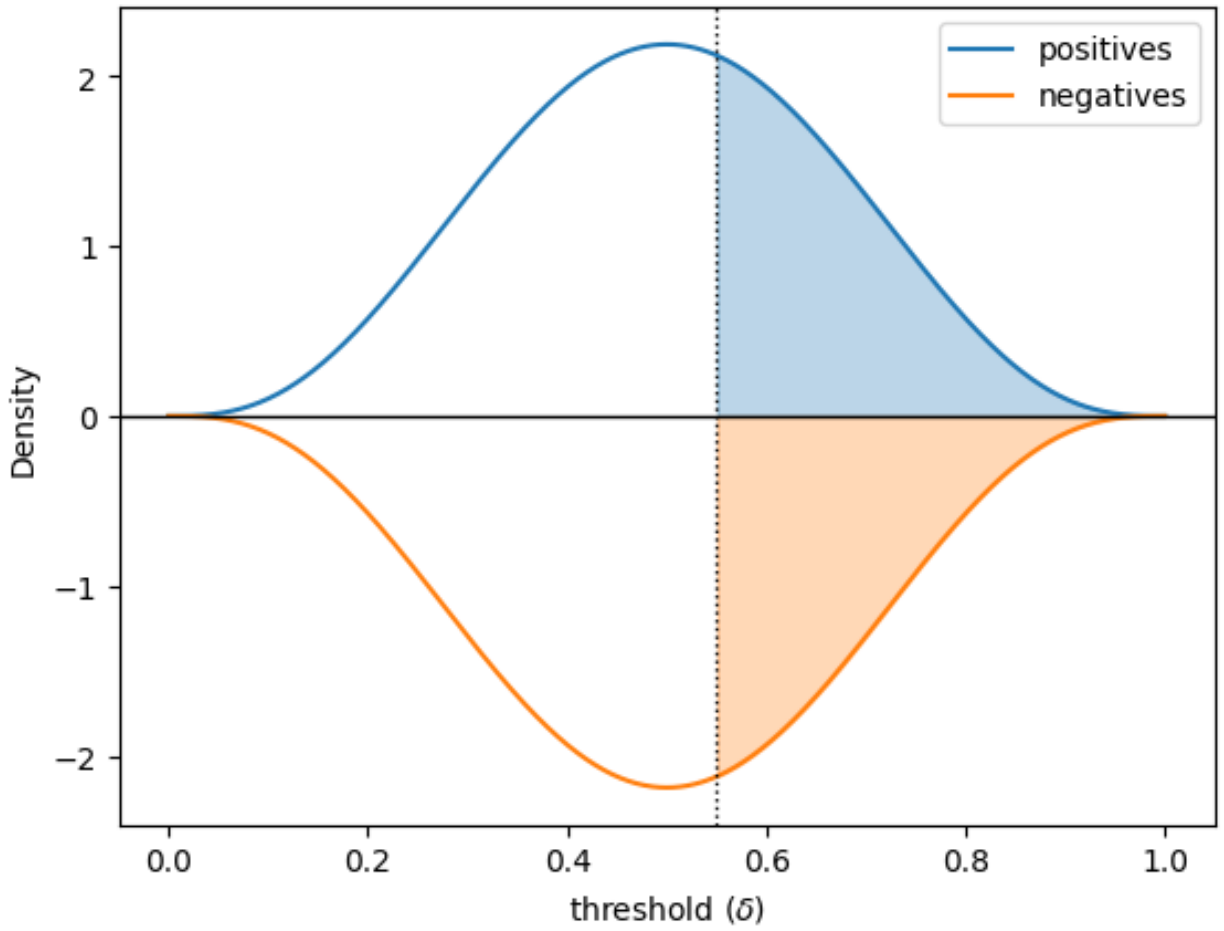
- Similar distributions between positive and negative cases.
- The predicted value tells you nothing about which one it's more likely to be.



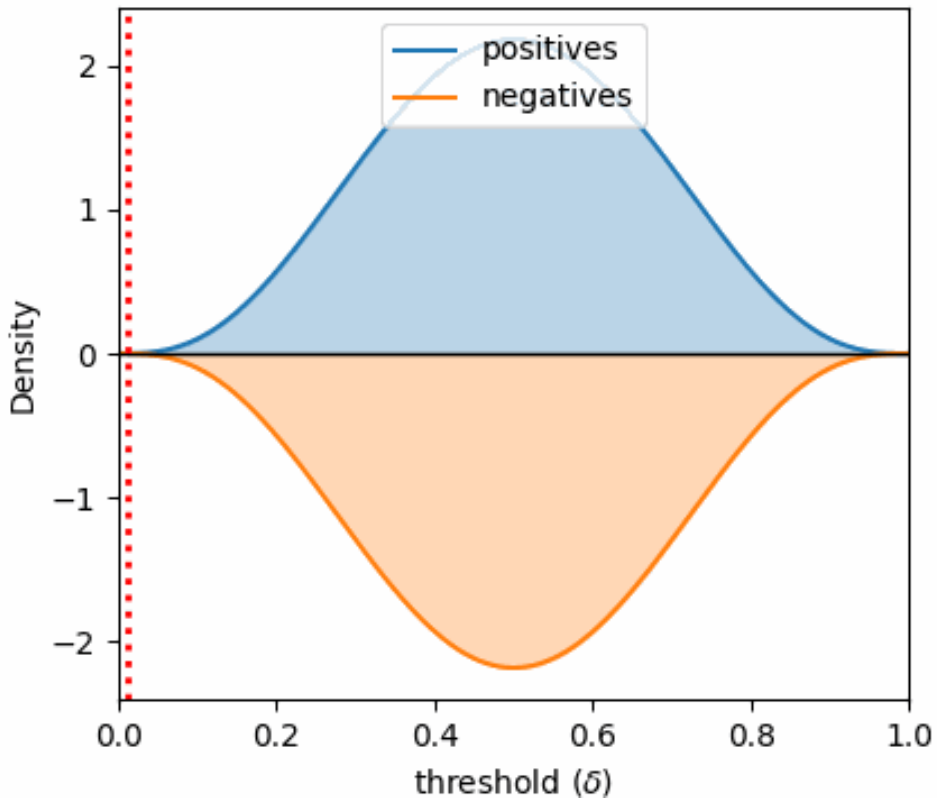
Suppose our features contain no information about the outcome. What might predictions look like?

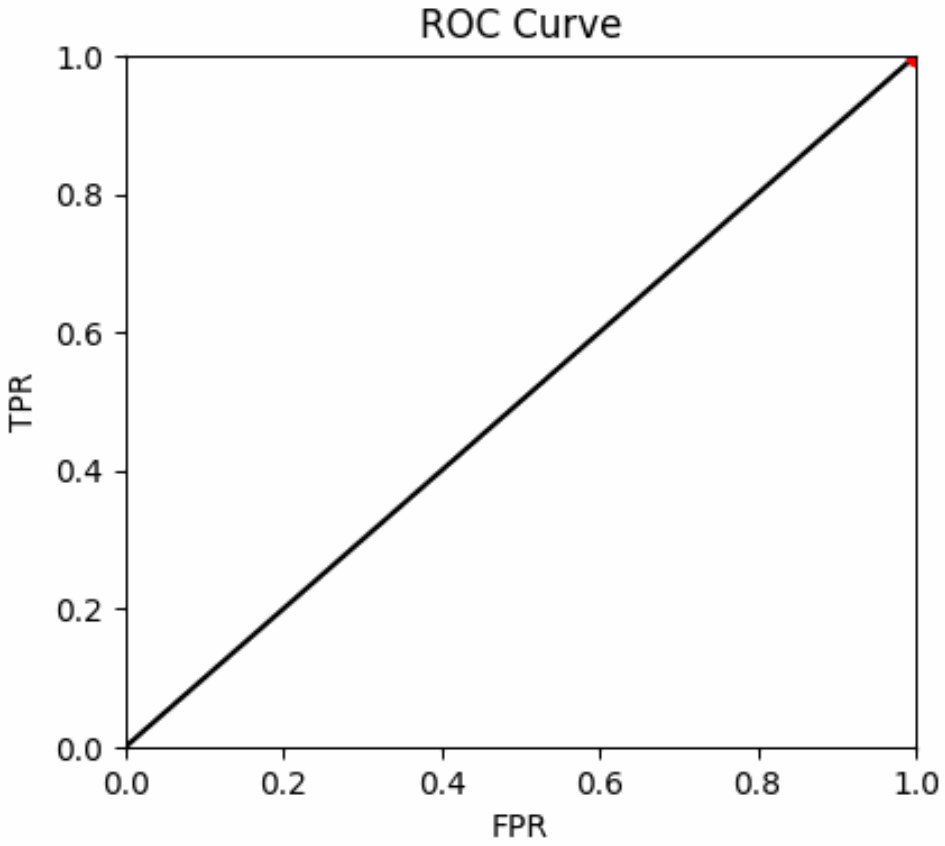
What is the:

(a) TPR (sensitivity)?
(b) FPR (1 – specificity)?
(c) PPV?



Our “no information” predictive model



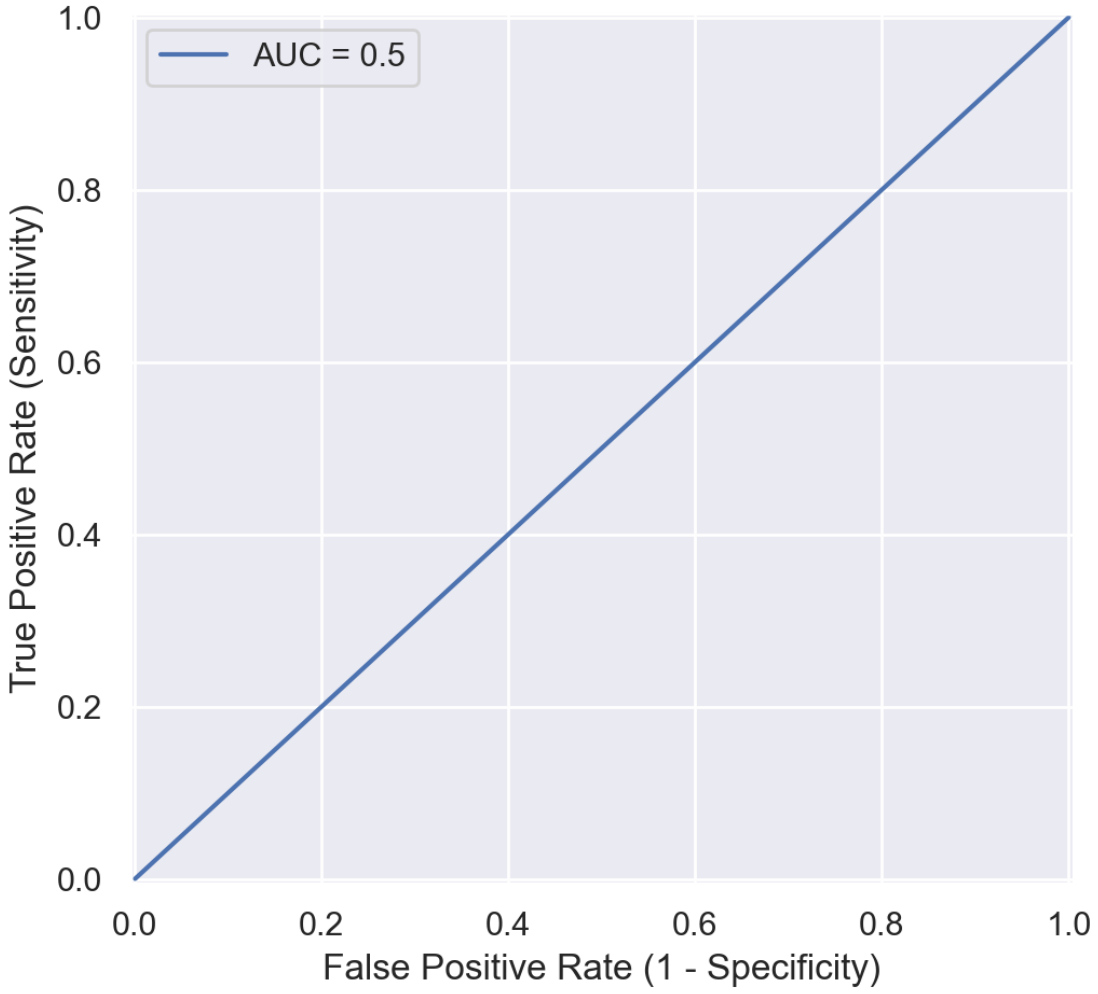


TPR and FPR always match (why?)

- the distributions of positive and negative cases match
- the corresponding CDFs match

Our “no information” model

- AUC is 0.5
- Do we ever see AUC < 0.5?
- If so, what can we infer about the distributions of our predictions for positive and negative samples?



So, what's a *good* AUC value?
(i.e., *good* performance)?

It depends.

- Are predictions better than random?
- Are predictions than the previous best performing model?
- Are predictions better than expert performance?
- Does performance exceed our (informed) expectations?

- **<u>Is the model clinically useful?</u>**

- <u>Different expectations for risk prediction vs classification</u>

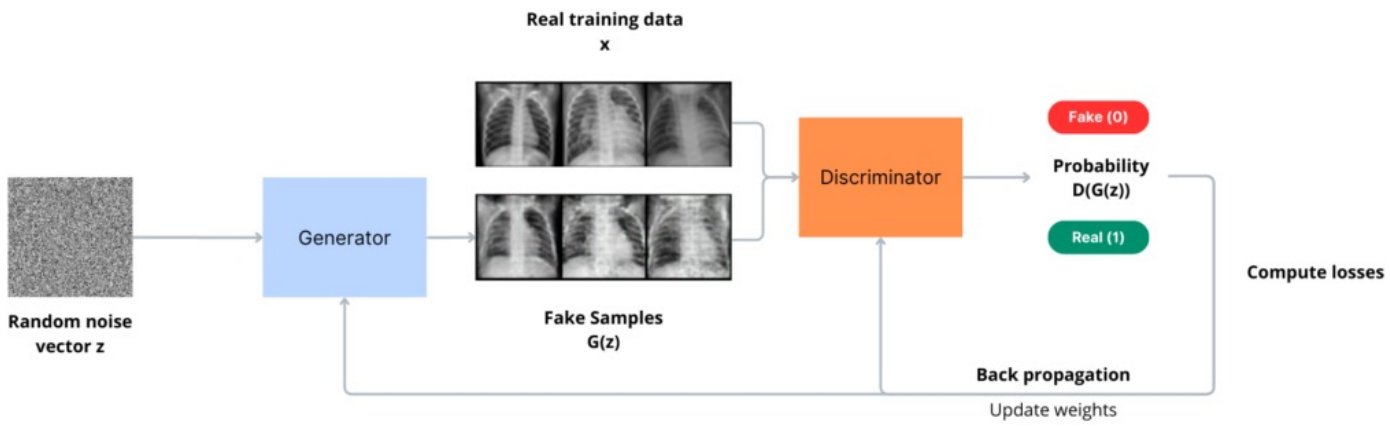
What can we say about prediction performance when distributions match?

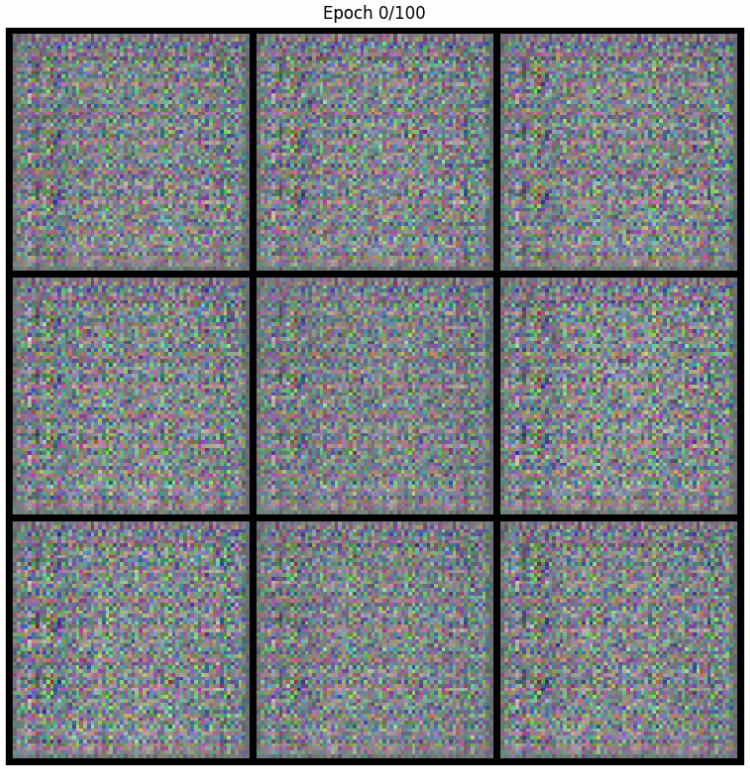
- When distributions of <u>predictions</u> match between pos/neg:
  - predictions provide no information about the outcome
  - i.e. random guessing

- When distributions of <u>features</u> (observed or latent) match between pos/neg:
  - features contain no information about the outcome

- Note that adversarial methods operate on this principle:
  - Train a *discriminator* that predicts pos/neg (or real/fake)
  - Train a *generator* whose goal is to <u>increase</u> the discriminator's loss (many variations on this)
  - Result? Distribution matching

Brief overview of generative adversarial networks (GANs)





$$\min_G \max_D V(D,G) = \mathbb{E}_{\boldsymbol{x}\sim p_{\text{data}}(\boldsymbol{x})}[\log D(\boldsymbol{x})] + \mathbb{E}_{\boldsymbol{z}\sim p_{\boldsymbol{z}}(\boldsymbol{z})}[\log(1 - D(G(\boldsymbol{z})))]$$

Taken from: https://blog.ovhcloud.com/understanding-image-generation-beginner-guide-generative-adversarial-networks-gan/

We've quantified performance across all thresholds, but how do we use the model?

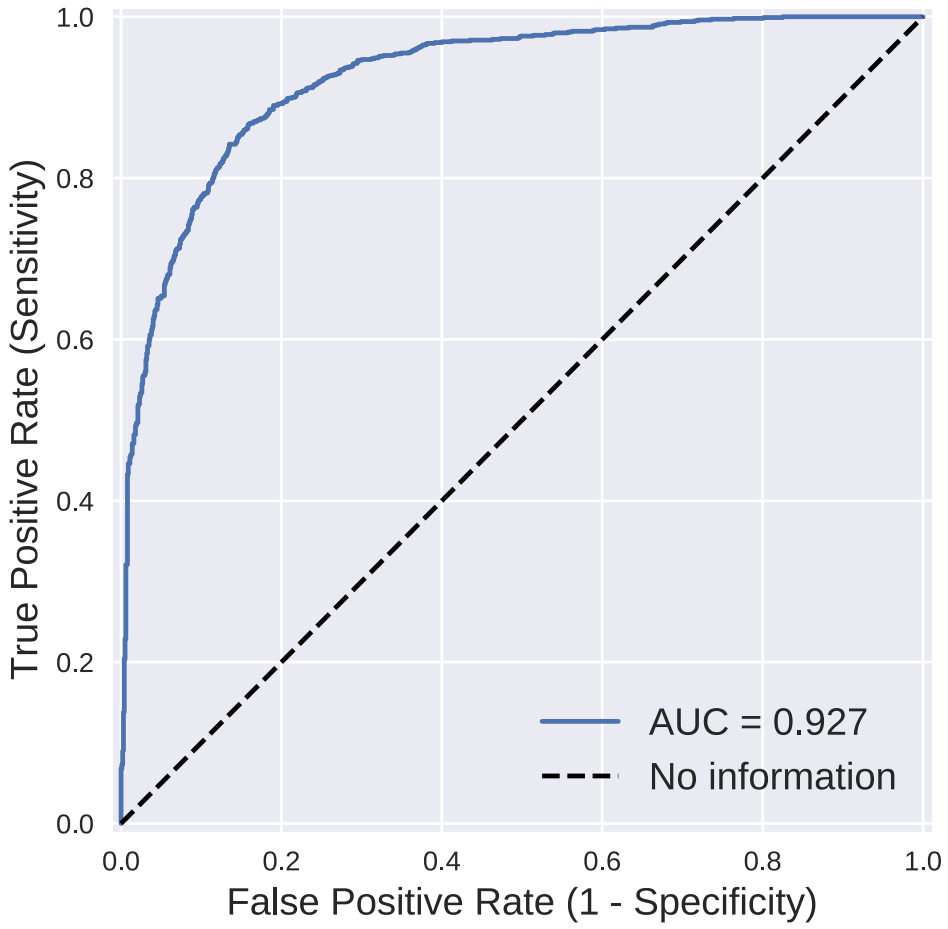
Sometimes the predicted probability really is what we care about.

- *Example*: probability of heart attack
- If so, we need to make sure our model is *calibrated*

More often, we need to pick a threshold so we can decide whether to:

- Alert a provider
- Get a biopsy
- Refer the patient
- etc

What threshold should we pick? What's the right tradeoff?



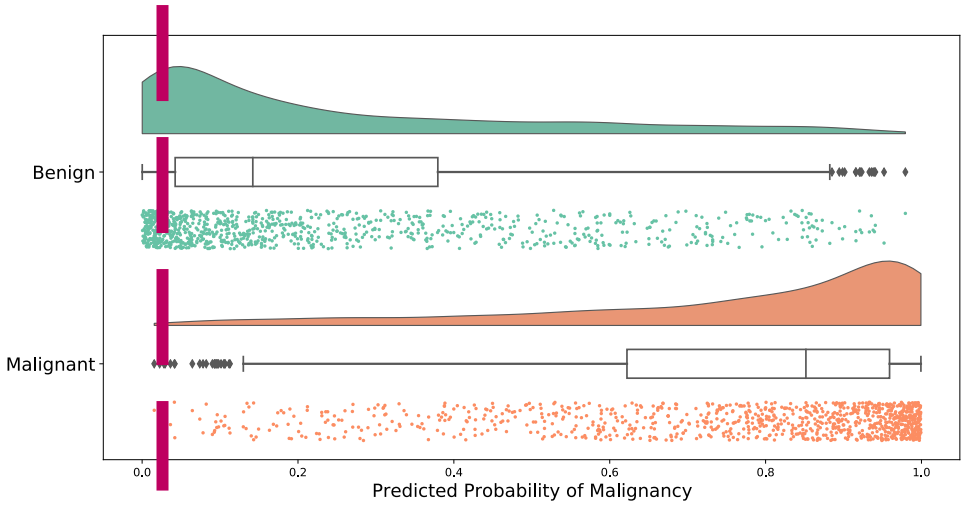
Healthcare Scenarios

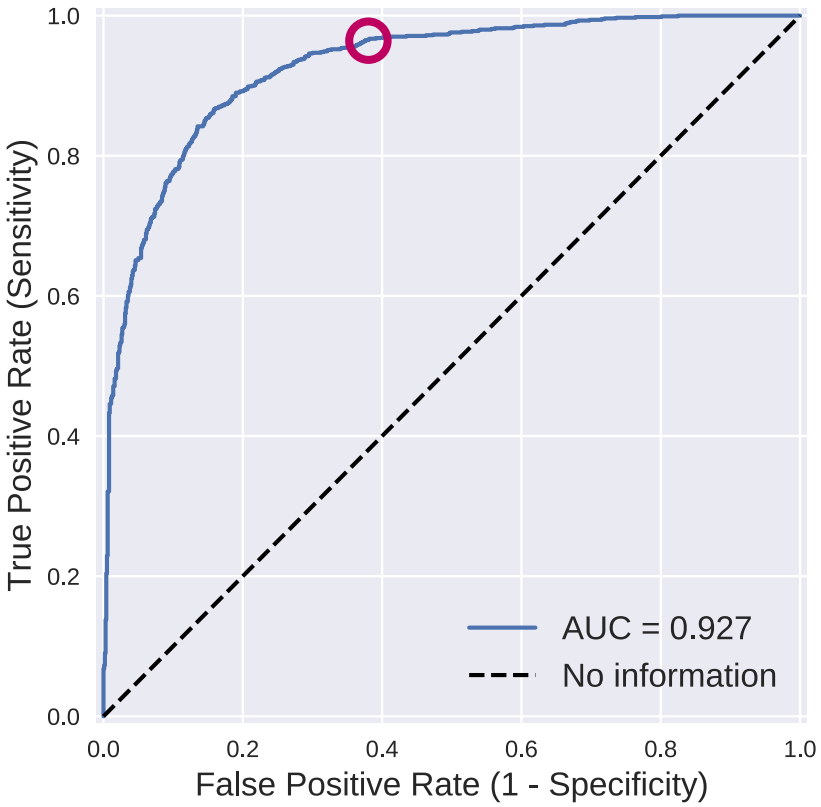
Which performance metric is most important?

1. A computer vision model that detects carcinoma

Operating Point:

high sensitivity

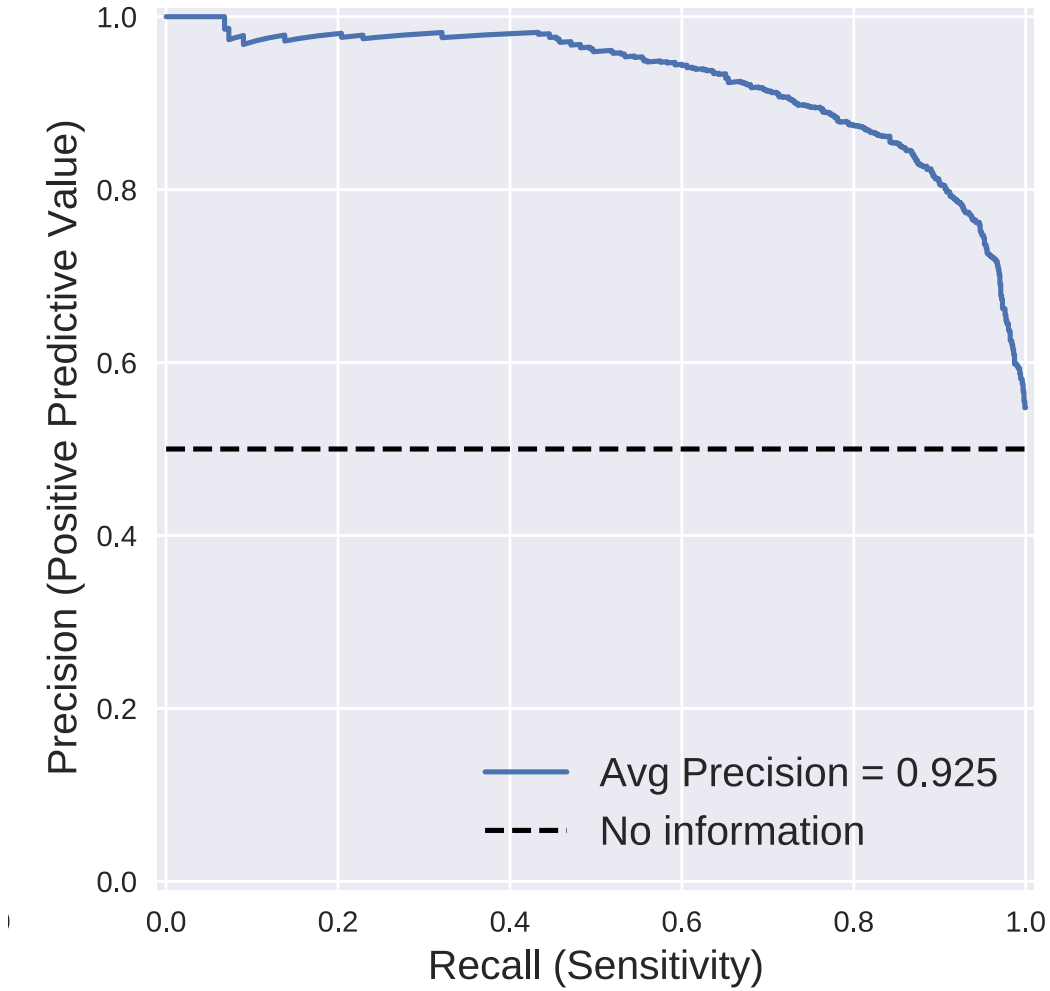




The Precision-Recall Curve

(i.e., PPV-Sensitivity Curve)

- Some argue it has greater clinical relevance than the ROC curve
- The *no information* classifier always achieves PPV equal to the *base rate*, or *prevalence* (why?)
- PPV as well as the area under this curve (average precision) must be interpreted relative to prevalence



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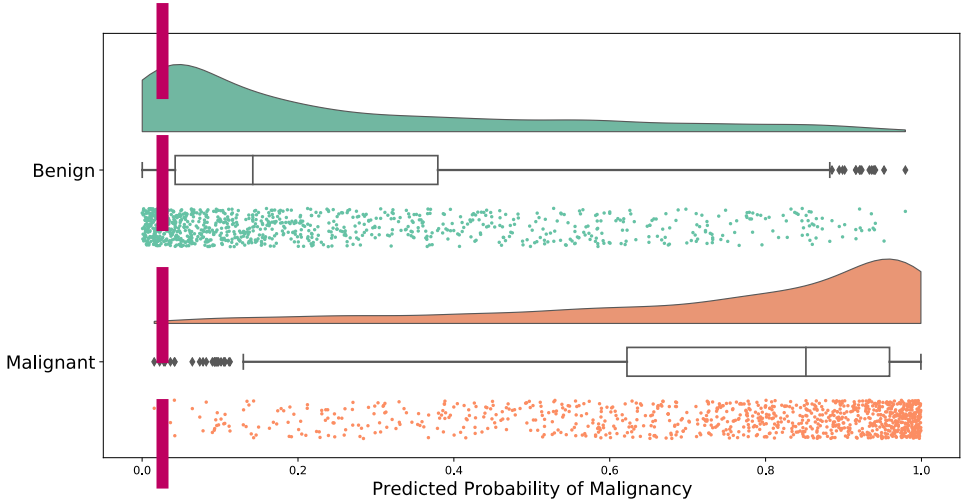
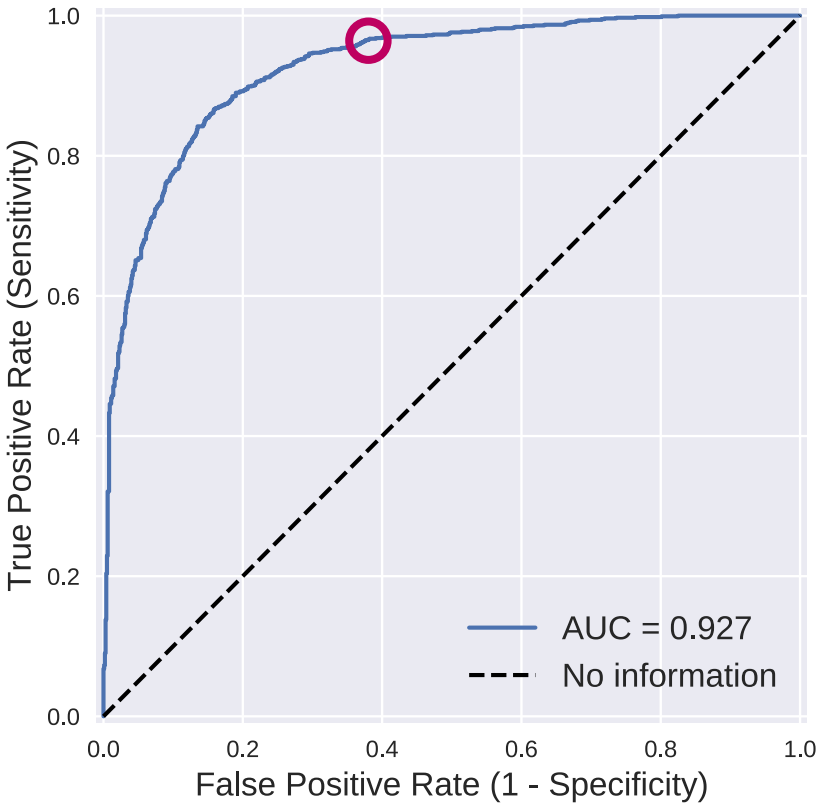
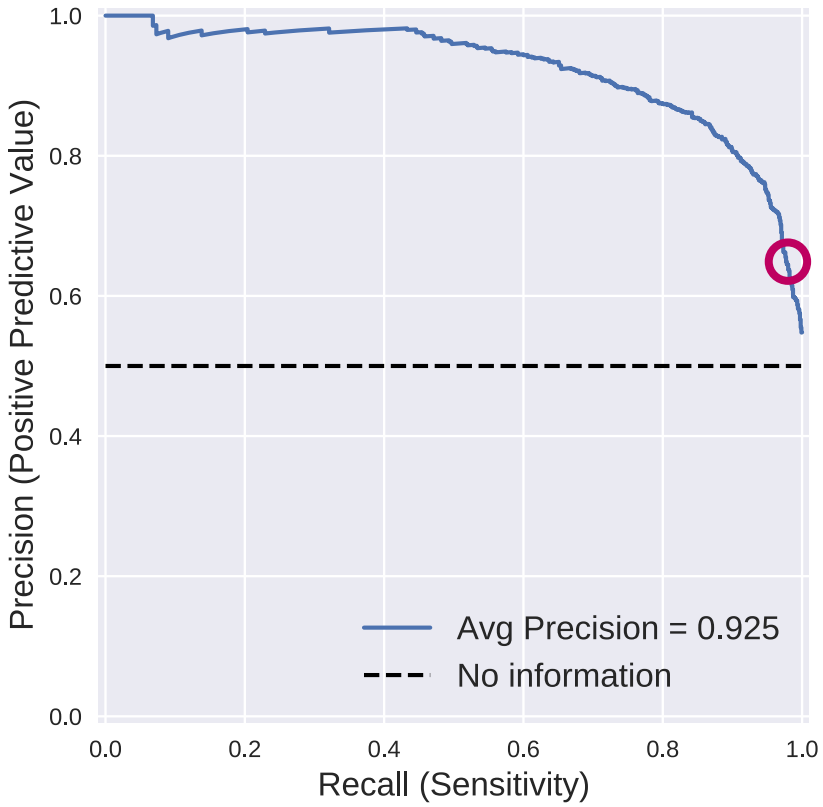
high sensitivity







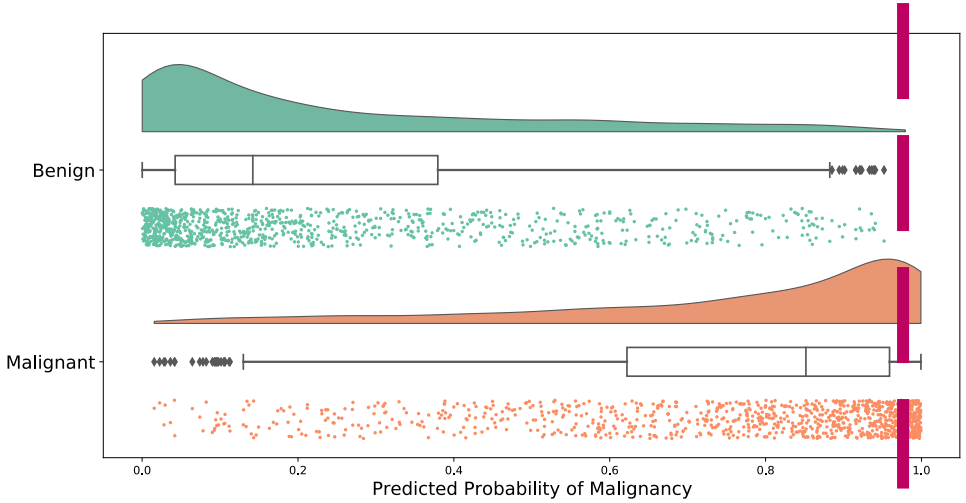
Healthcare Scenarios

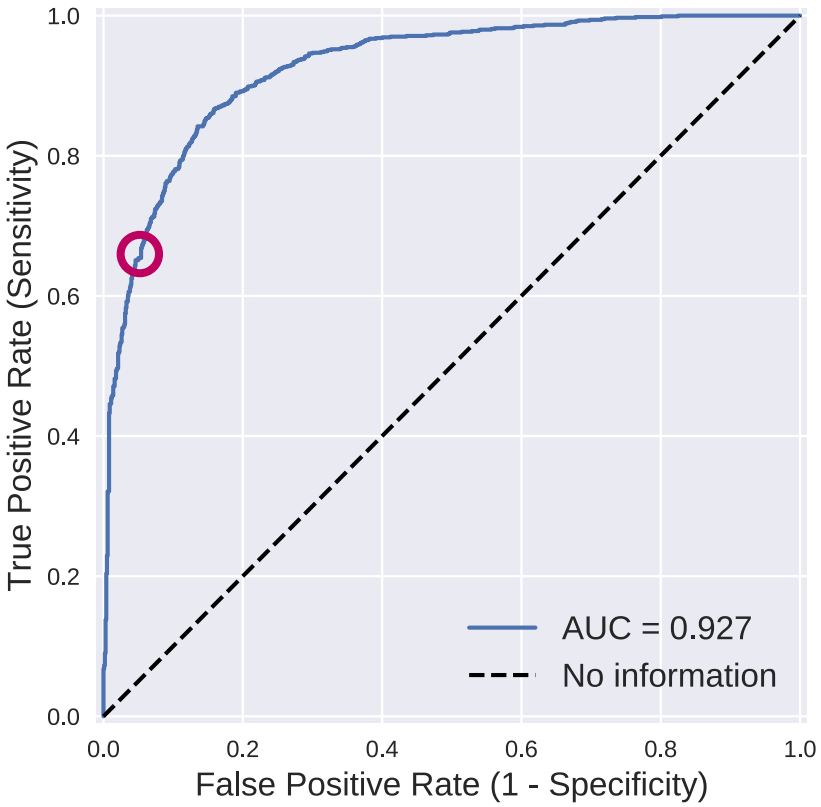
1. A computer vision model that detects carcinoma

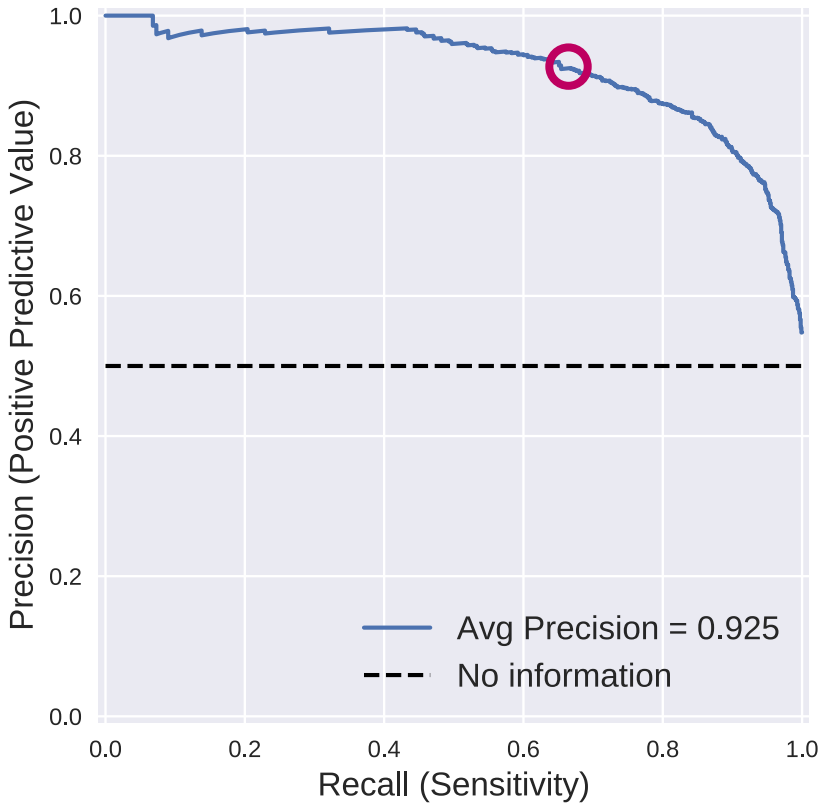
2. An algorithm that detects atrial fibrillation in Apple Watch users

Operating Point:

high specificity







Healthcare Scenarios

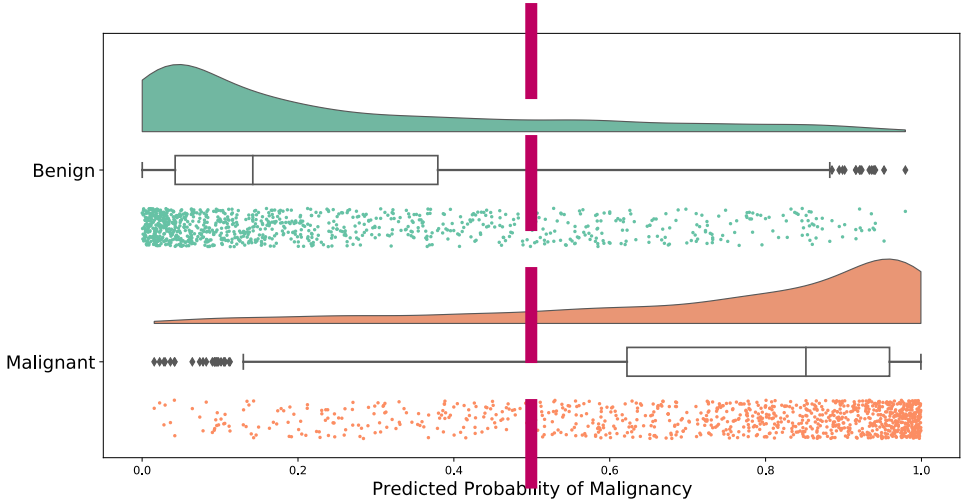
1. A computer vision model that detects carcinoma

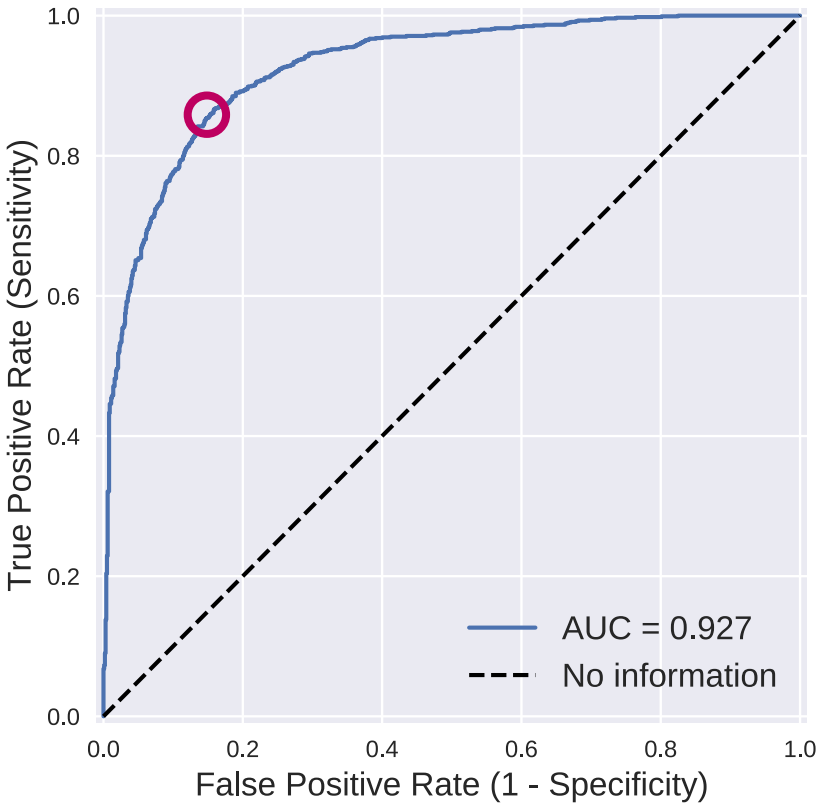
2. An algorithm that detects atrial fibrillation in Apple Watch users

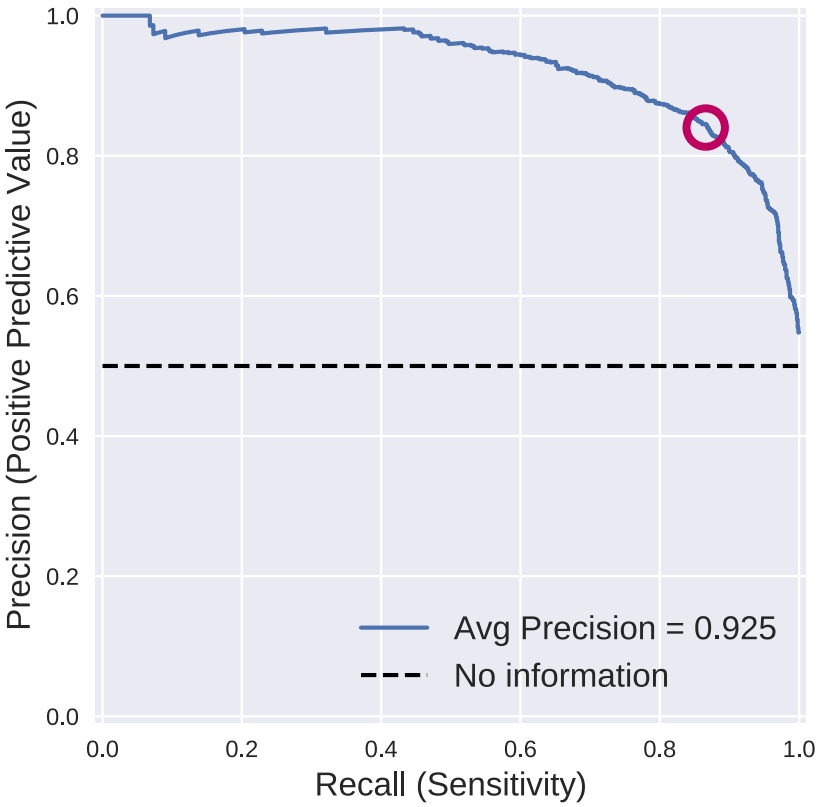
3. An EHR-based model that monitors autism risk

Operating Point:

balanced







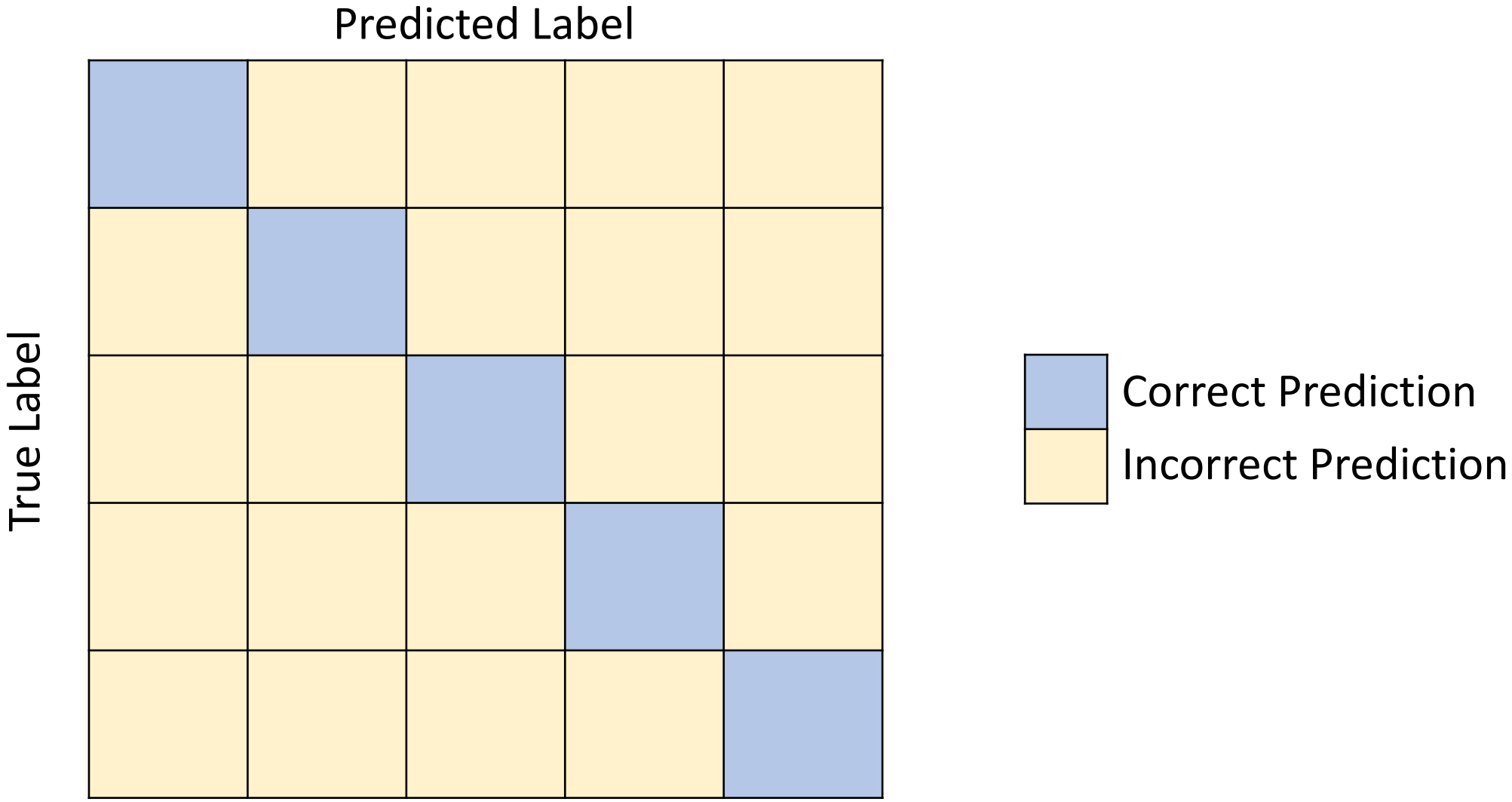
Healthcare Scenarios

1. A computer vision model that detects carcinoma
2. An algorithm that detects atrial fibrillation in Apple Watch users
3. An EHR-based model that monitors autism risk

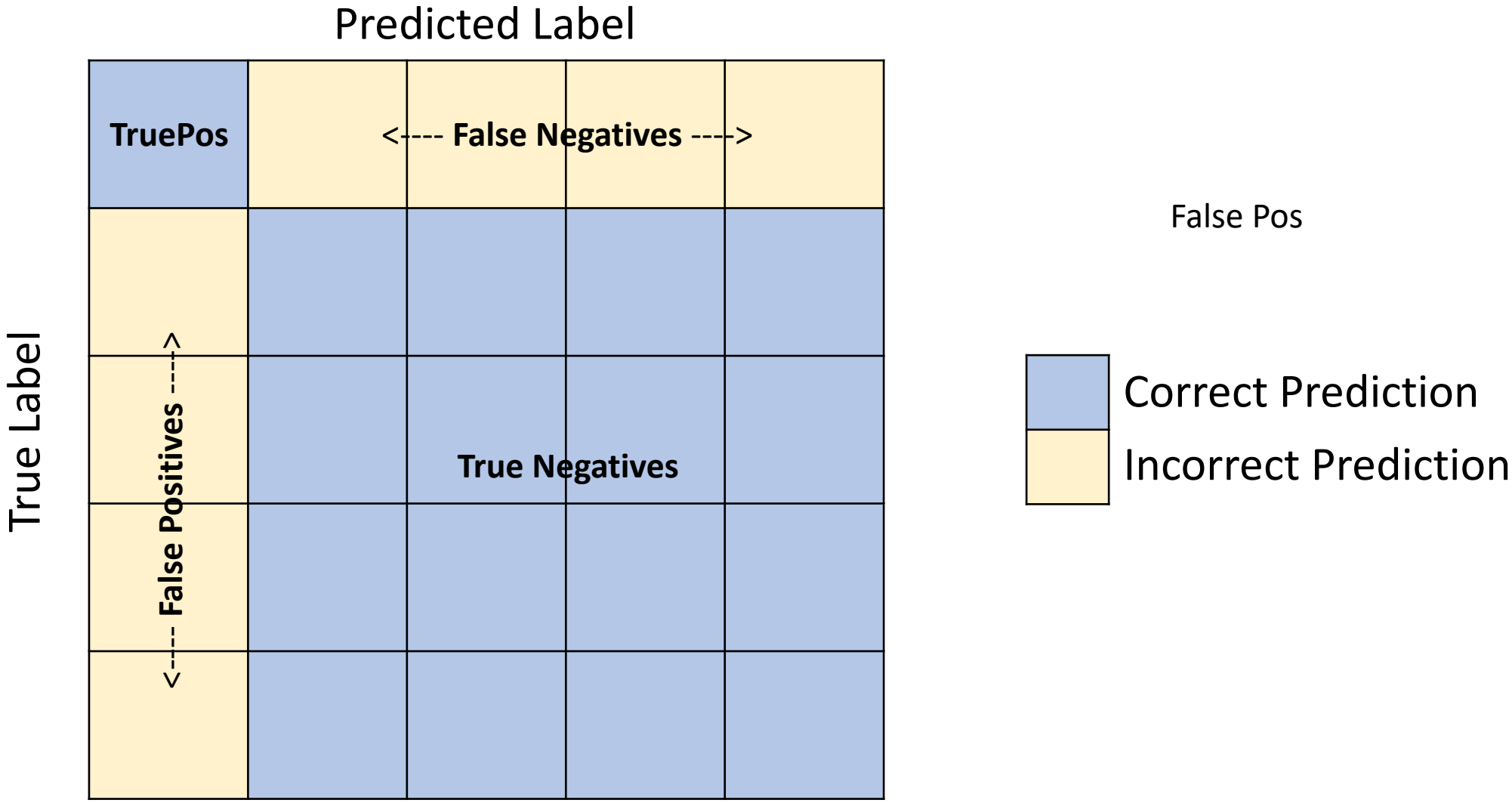
- Sometimes specificity and sensitivity are difficult to interpret, particularly for rare conditions or events.
- The most clinically relevant measure is often the positive predictive value (or negative predictive value).

Multi-class problems

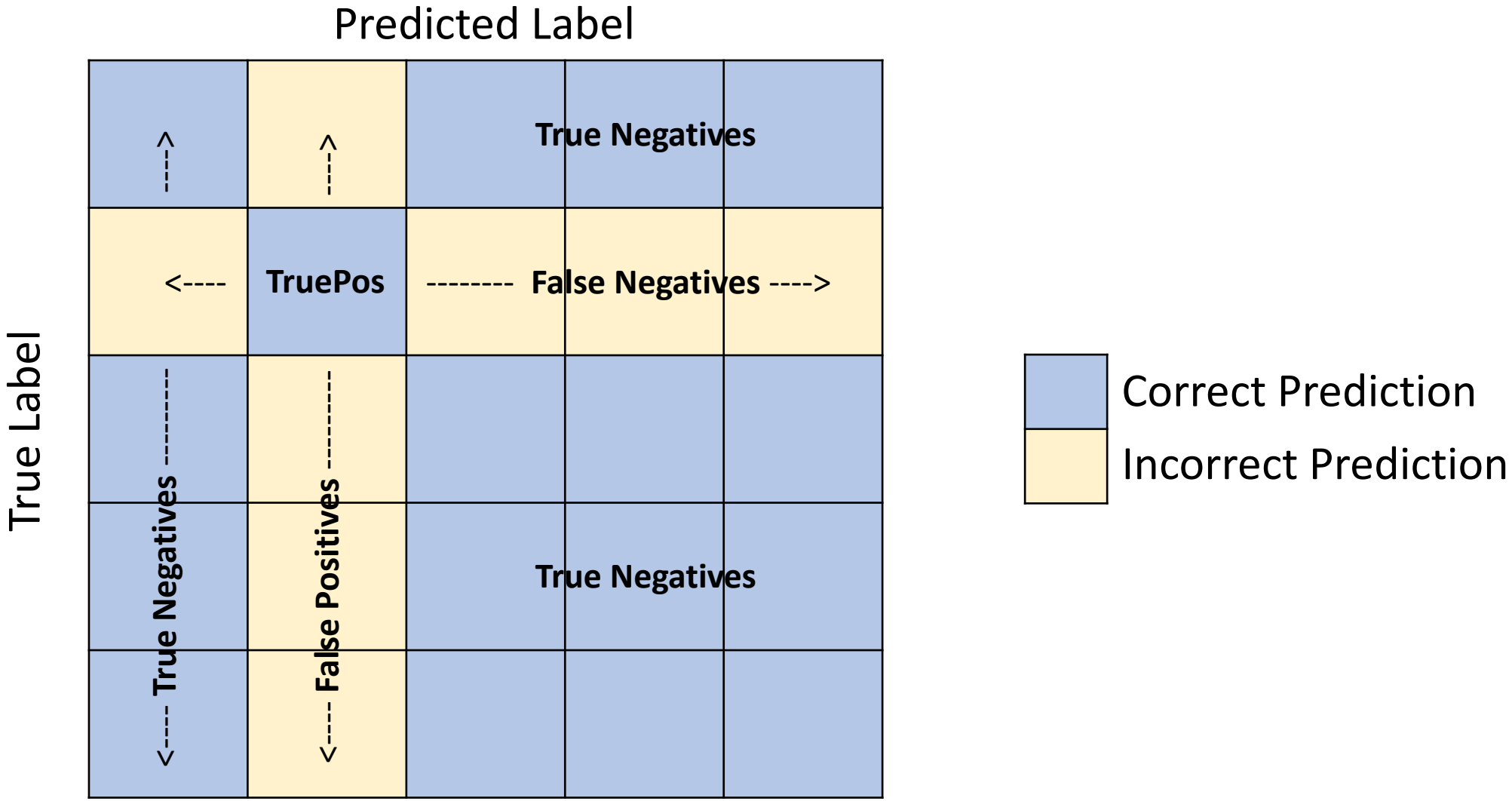
Multi-class problems: “Confusion Matrix”



Multi-class problems: Binary for Label 1



Multi-class problems: Binary for Label 2



Prediction for Ms. Z

After deciding to use the model, you see an 82 year-old woman with a new diagnosis of diastolic heart failure after a recent hospitalization. She had an echo and a cath showing left ventricular hypertrophy and grade 2 diastolic dysfunction with a preserved ejection fraction. She has a history of diabetes and retinopathy and neuropathy.

The model predicts that she is 25% likely (95th percentile) to progress to stage IV CKD within the next 2 years. This prediction is consistent with your clinical impression, so you consider whether to refer her to Nephrology.

Should you refer her? Why or why not?

What level of risk justifies referral?

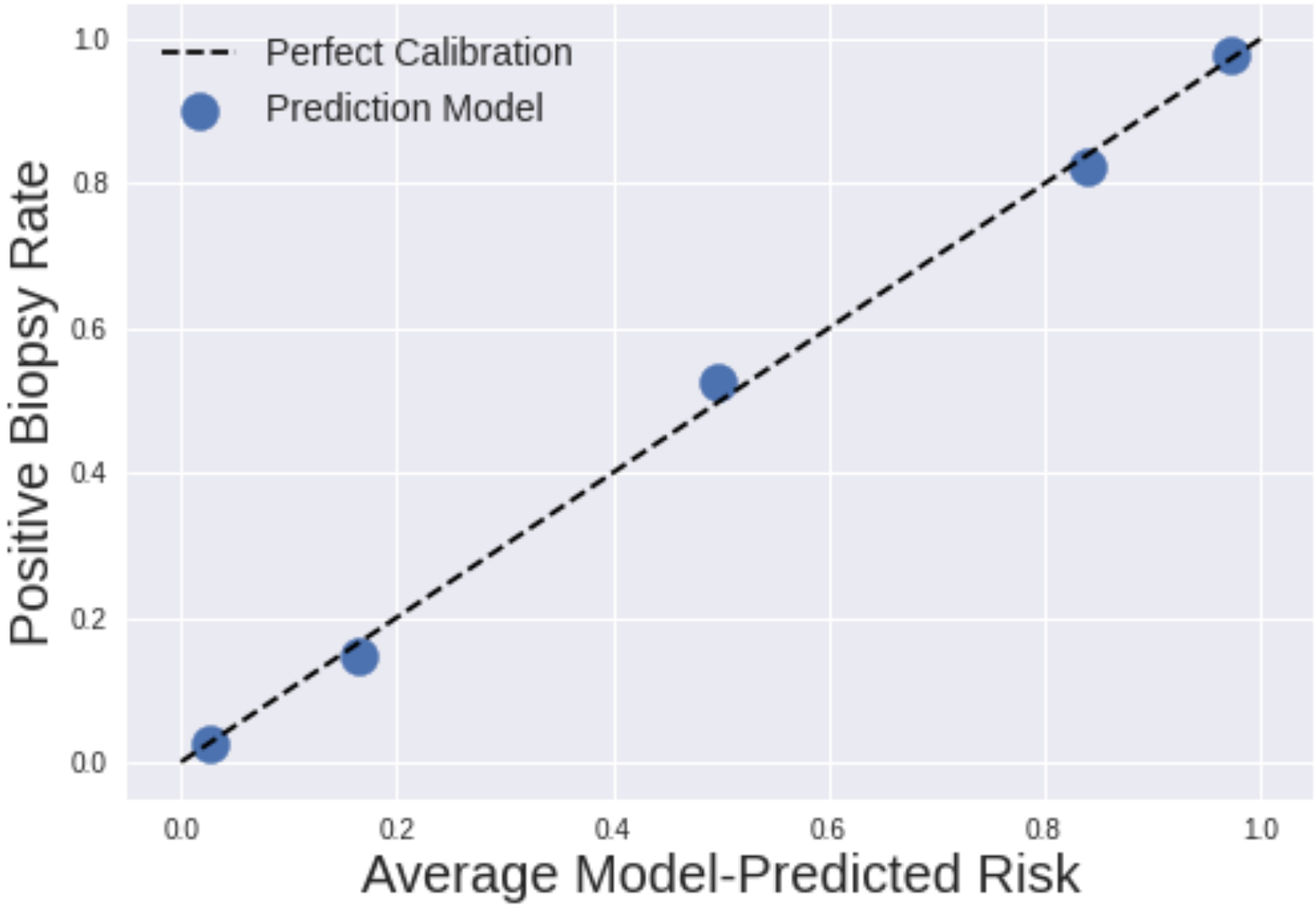
Ms. Z's case leads you to wonder whether referral would be appropriate in more borderline cases.

To make the process more streamlined and consistent, you decide you'd like to refer patients whose risk is above a particular threshold.

- How do you decide on this threshold?
- Should it be the same for everyone?
- What does your decision say about cost/benefit?

Assessing calibration

Assess Calibration Graphically



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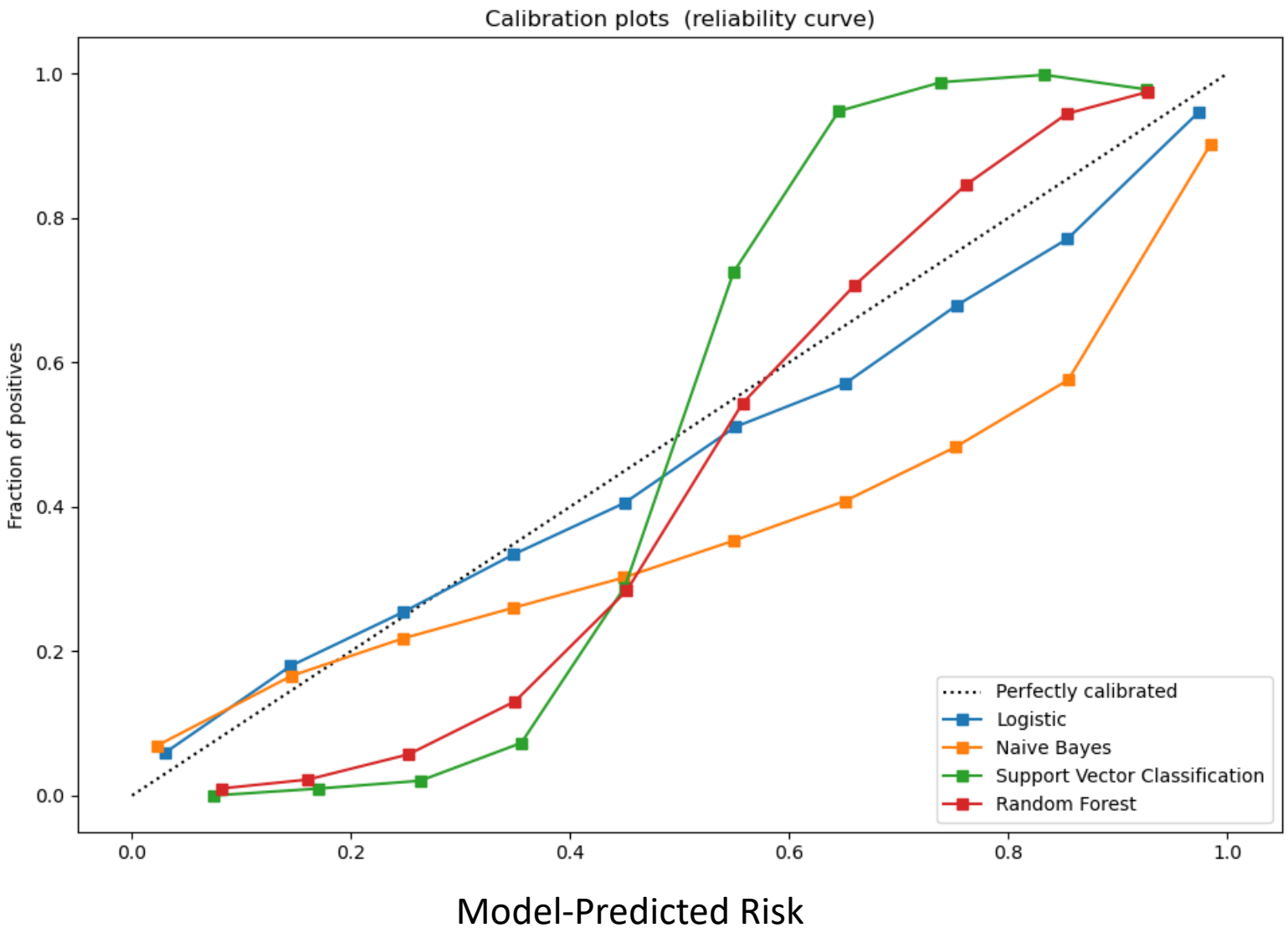

From: https://scikit-learn.org/stable/modules/calibration.html

Investigating model predictions

Looking inside the black box

Prediction for Mr. X

You've been using your stage IV prediction model for several months now and have found its predictions to be helpful and in line with your clinical judgment.

You see a 32 year-old male patient with new significant hypertension (165/91). You check your model and see it predicts low stage IV CKD risk (<2%). However, on physical exam there is puffiness around his eyes and 1+ edema to his mid-shin. On urinalysis you find significant proteinuria (3+).

The model has been reliable thus far, and you wish to reconcile the findings you're seeing with its predictions.

<u>Report Back</u>: What can you do to investigate?

Key Questions

- What information is available to the model?
  - Predictor groups
  - Timing of collection
  - Lab *ID* or lab *values*?

- What population was used to train the model?
  - Relevant inclusion / exclusion criteria
  - Representation of specific demographics

- <u>What do you know that the model doesn't (or vice versa)?</u>

Report Back:
What else should you know to use the model?

Model Facts **Model name:** Deep Sepsis **Locale:** Duke University Hospital

Approval Date: 09/22/2019 **Last Update:** 01/13/2020 **Version:** 1.0

Summary

This model uses EHR input data collected from a patient's current inpatient encounter to estimate the probability that the patient will meet sepsis criteria within the next 4 hours. It was developed in 2016-2019 by the Duke Institute for Health Innovation. The model was licensed to Cohere Med in July 2019.

Mechanism

- **Outcome** ……sepsis within the next 4 hours, see outcome definition in "Other Information"
- **Output** ……0% - 100% probability of sepsis occurring in the next 4 hours
- **Target population** ……all adult patients >18 y.o. presenting to DUH ED
- **Time of prediction** ……every hour of a patient's encounter
- **Input data source**……electronic health record (EHR)
- **Input data type** ……demographics, analytes, vitals, medication administrations
- **Training data location and time-period** ……DUH, diagnostic cohort, 10/2014 – 12/2015
- **Model type**…… Recurrent Neural Network

Validation and performance

	Prevalence	AUC	PPV @ Sensitivity of 60%	Sensitivity @ PPV of 20%	Cohort Type	Cohort URL / DOI
Local Retrospective	18.9%	0.88	0.14	0.50	Diagnostic	arxiv.org/abs/1708.05894
Local Temporal	6.4%	0.94	0.20	0.66	Diagnostic	jmir.org/preprint/15182
Local Prospective	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**
External	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**
Target Population	6.4%	0.94	0.20	0.66	Diagnostic	jmir.org/preprint/15182

Uses and directions

- **Benefits:** Early identification and prompt treatment of sepsis can improve patient morbidity and mortality.
- **Target population and use case:** Every hour, data is pulled from the EHR to calculate risk of sepsis for every patient at the DUH ED. A rapid response team nurse reviews every high-risk patient with a physician in the ED to confirm whether or not to initiate treatment for sepsis.
- **General use:** This model is intended to be used to by clinicians to identify patients for further assessment for sepsis. The model is not a diagnostic for sepsis and is not meant to guide or drive clinical care. This model is intended to complement other pieces of patient information related to sepsis as well as a physical evaluation to determine the need for sepsis treatment.
- **Appropriate decision support:** The model identifies patient X as at a high risk of sepsis. A rapid response team nurse discusses the patient with the ED physician caring for the patient and they agree the patient does not require treatment for sepsis.
- **Before using this model:** Test the model retrospectively and prospectively on a diagnostic cohort that reflects the target population that the model will be used upon to confirm validity of the model within a local setting.
- **Safety and efficacy evaluation:** Analysis of data from clinical trial (NCT03655626) is underway. Preliminary data shows rapid response team, nurse-driven workflow was effective at improving sepsis treatment bundle compliance.

Warnings

- **Risks:** Even if used appropriately, clinicians using this model can misdiagnose sepsis. Delays in a sepsis diagnosis can lead to morbidity and mortality. Patients who are incorrectly treated for sepsis can be exposed to risks associated with unnecessary antibiotics and intravenous fluids.
- **Inappropriate Settings:** This model was not trained or evaluated on patients receiving care in the ICU. Do not use this model in the ICU setting without further evaluation. This model was trained to identify the first episode of sepsis during an inpatient encounter. Do not use this model after an initial sepsis episode without further evaluation.
- **Clinical Rationale:** The model is not interpretable and does not provide rationale for high risk scores. Clinical end users are expected to place model output in context with other clinical information to make final determination of diagnosis.
- **Inappropriate decision support:** This model may not be accurate outside of the target population, primarily adults in the non-ICU setting. This model is not a diagnostic and is not designed to guide clinical diagnosis and treatment for sepsis.
- **Generalizability:** This model was primarily evaluated within the local setting of Duke University Hospital. Do not use this model in an external setting without further evaluation.
- **Discontinue use if:** Clinical staff raise concerns about utility of the model for the indicated use case or large, systematic changes occur at the data level that necessitates re-training of the model.

Other information:

- **Outcome Definition:** https://doi.org/10.1101/648907
- **Related model:** http://doi.org/10.1001/jama.2016.0288
- **Model development & validation:** arxiv.org/abs/1708.05894
- **Model implementation**: jmir.org/preprint/15182
- **Clinical trial:** clinicaltrials.gov/ct2/show/NCT03655626
- **Clinical impact evaluation:** TBD
- **For inquiries and additional information:** please email mark.sendak@duke.edu

Sendak, M.P., Gao, M., Brajer, N. *et al.* Presenting machine learning model information to clinical end users with model facts labels. *npj Digit. Med.* **3**, 41 (2020). https://doi.org/10.1038/s41746-020-0253-3